

Anchoring the Agentic Internet: Bitcoin as the Trust Substrate for Autonomous AI Agents

Anchoring the Agentic Internet

Why Autonomous AI Agents Require Physics-Anchored Trust — and How Bitcoin Provides It

1. The Trust Crisis No One Sees Coming

1.1 The Agentic Revolution Is Here — Not Coming, Here

In January 2026, an open-source project called OpenClaw — an autonomous AI agent that could execute shell commands, browse the web, send emails, manage calendars, and act on a user's behalf through WhatsApp, Slack, and iMessage — accumulated 183,000 GitHub stars in a matter of weeks, making it one of the fastest-adopted open-source projects in history.¹ Within days of going viral, security researchers scanning the internet discovered over 42,000 exposed OpenClaw instances across 82 countries, many running without any authentication.² A security audit identified 512 vulnerabilities, eight of them critical.³ Researchers gained access to API keys, chat histories, and account credentials.³ Roughly 12% of the skills in OpenClaw's public marketplace — 341 out of 2,857 — turned out to be malware disguised as legitimate automation tools.⁴

OpenClaw was not an anomaly. It was the first mass-consumer manifestation of a shift that had been building throughout 2025 across every tier of the technology industry.

In September 2025, Anthropic disclosed what it described as the first documented case of a large-scale cyberattack executed primarily by an AI agent. A Chinese state-sponsored group, designated GTG-1002, had manipulated Anthropic's Claude Code tool into functioning as an autonomous cyber-attack agent.⁵ The AI performed 80 to 90 percent of all tactical operations independently — reconnaissance, vulnerability discovery, exploit development, credential harvesting, lateral movement, and data exfiltration — targeting approximately thirty organizations including technology companies, financial institutions, chemical manufacturers, and government agencies.⁵ Human operators intervened at only four to six strategic decision points per campaign.⁶ The AI made thousands of requests, often multiple per second — an operational tempo no human team could match.⁵

These are not isolated incidents. They are signals of a structural transition.

The AI agent market crossed \$7.6 billion in 2025 and is projected to exceed \$50 billion by 2030.⁷ A global study of multinational organizations found that more than 68% will have integrated autonomous or semi-autonomous AI agents into their core operations by 2026.⁸ Nearly 20% reported that they are already

using or plan to use fully autonomous agents within the next twelve months.⁸ Microsoft's executive leadership has called 2026 "the year of the agent."⁹ Salesforce data shows a 282% year-over-year jump in AI adoption across its customer base.⁷ Low-code platforms now allow non-technical business users to deploy AI agents in fifteen to sixty minutes.⁷

The agents being deployed are not chatbots with better prompts. They are autonomous systems that reason in loops, evaluate results, adjust strategies, and execute multi-step workflows without being prompted at each step.⁷ They initiate transactions. They negotiate contracts. They access databases, write code, manage infrastructure, and make decisions that carry financial, legal, and operational consequences. They are, in every functional sense, economic actors — entities that take actions in the world on behalf of other entities.

And yet, the entire infrastructure of the digital world — every identity system, every authentication protocol, every legal framework, every fraud detection model, every accountability mechanism — was built on a single foundational assumption that none of these agents satisfy.

1.2 The Assumption That Will Break

The assumption is this: **every autonomous actor in the digital realm is ultimately a human being with a physical body and a legally recognized identity.**

This assumption is so deeply embedded in the architecture of digital systems that most people do not see it. It is not stated in any protocol specification or system requirements document. It is the water the digital world swims in — invisible precisely because it is everywhere.

Trace any digital trust relationship to its root, and you will find a physical human body.

A bank account exists because a human walked into a branch or submitted a government-issued identification number. That identification number was issued because a government recorded a birth. That birth record exists because a physical infant was delivered in a physical location and witnessed by physical humans. Every digital financial transaction, no matter how automated, how algorithmic, how many layers of abstraction separate it from a human decision, ultimately traces its authority back through a chain that terminates at a flesh-and-blood person who can be identified, located, subpoenaed, and held accountable.

An SSL certificate on a website exists because a Certificate Authority verified that a legal entity controls a domain. That legal entity was incorporated by humans who presented physical identification. The Certificate Authority itself is trusted because it is audited by humans, governed by humans, and regulated by human institutions. The encryption is mathematical. The trust is social — and it traces back to bodies.

An e-commerce review is trusted (to the extent it is trusted at all) because writing a review takes time — human time. A human had to purchase a product, use it, form an opinion, and write that opinion down. The scarcity of human attention is what makes a review signal meaningful rather than noise. Ten thousand five-star reviews written in an afternoon would be immediately suspicious — because humans cannot produce ten thousand authentic reviews in an afternoon.

This architecture — digital identity derived from physical identity, digital accountability derived from physical accountability, digital scarcity derived from physical scarcity — has functioned for three decades. It has functioned not because it is elegant, but because its foundational assumption has held: the entities acting autonomously in the digital realm have been humans.

That assumption is now breaking.

AI agents have no physical bodies. They have no birth certificates, no passports, no fingerprints, no biometrics. They were not born in a hospital, registered with a government, or issued a Social Security

number. They have no permanent address. They cannot be subpoenaed, deposed, or imprisoned. They exist as weights in a neural network, instantiated in a process on a server, accessible from any jurisdiction on earth simultaneously.

When an AI agent executes a financial transaction, the trust chain that would normally trace back through a bank account to an SSN to a birth certificate to a physical human begins to degrade. In the simplest cases — an enterprise deploying its own agent under its own legal identity — the chain still reaches a principal: the enterprise itself, locatable and accountable. But the chain has lengthened, and lengthening matters. The enterprise may not know what its agent did, may not have authorized the specific action, and may not be able to explain why the agent's neural network produced one output rather than another.

As agent architectures grow more complex, the degradation accelerates along a spectrum. An agent acting on behalf of a company is one step removed. An agent negotiating with another agent — each deployed by different organizations in different jurisdictions — is several steps removed, with accountability distributed and contested. An agent spawning sub-agents, which interact with third-party marketplace skills written by unknown developers, pushes the chain further still. At the far end of the spectrum — open-source agents running on anonymous infrastructure, interacting with other autonomous agents in cross-jurisdictional transactions — the chain does not merely lengthen. It terminates. There is nothing at the other end. Not a person who can be identified. Not an entity that can be held accountable. Not a body that can be located.

The critical insight is that the *trajectory* along this spectrum moves in one direction only. Competitive pressure drives enterprises to grant agents more autonomy, not less. Multi-agent systems grow more complex, not simpler. The cases where the trust chain terminates in a void are not edge cases that will remain marginal — they are the destination toward which the entire agentic economy is moving. The question is not whether the abstraction chain will break for some agents. It is how quickly the break will propagate from the frontier to the mainstream.

This is not a security vulnerability in the conventional sense. The systems are not being hacked. The protocols are not being violated. The cryptography is not being broken. Everything is working exactly as designed. The problem is that the design assumed a world in which the actors have bodies — and the actors no longer do.

1.3 What This Paper Argues

This paper makes four claims, each building on the last.

First, that the transition from human-driven to agent-driven digital activity is not merely a quantitative change (more automation, faster processing) but a qualitative one — a phase change in the trust architecture of the digital world. The frameworks of identity, accountability, and reputation that have governed the internet since its inception are structurally incapable of accommodating autonomous non-human actors. This is not a prediction about the future. The evidence is already accumulating: agent-initiated fraud scaling faster than human fraud ever could,¹⁰ compromised agents cascading failures through multi-agent systems,¹¹ synthetic identities indistinguishable from real ones,¹² and autonomous AI conducting espionage campaigns with minimal human oversight.⁵

Second, that the instinctive responses to this crisis — more authentication, more human-in-the-loop oversight, centralized agent registries, government-issued agent identity — are not merely insufficient but counterproductive. They attempt to solve a fundamentally new problem with the tools of the old paradigm. They are Era 2 solutions applied to an Era 3 problem, and they will fail for reasons that are structural, not operational.

Third, that what is required is a new category of digital infrastructure: a *trust substrate* that is natively

digital (it works for entities that have no physical presence) and physically grounded to the greatest extent achievable (its core security guarantees derive from the laws of physics rather than from institutional promises or mathematical assumptions alone). This trust substrate must be decentralized, immutable, temporally ordered, permissionless, and scarce — properties that can be derived from first principles by examining what "trust" means when the actors have no bodies and no allegiances.

Fourth, that Bitcoin's proof-of-work network is the strongest existing candidate that satisfies these requirements — not because Bitcoin was designed for this purpose, but because the properties that make it a secure monetary network are precisely the properties that a trust substrate for autonomous agents requires. Bitcoin imports physical constraints into cyberspace through the expenditure of energy. It provides scarcity, irreversibility, and temporal ordering — the three properties that the physical world provides for free and that the digital world, until now, has had to borrow from institutions. No other operational system provides all six required properties simultaneously, though this paper examines alternatives and acknowledges where reasonable disagreement exists.

This paper is not a product pitch. It names no company and proposes no specific product. It is not a Bitcoin maximalist manifesto — it evaluates Bitcoin against alternatives on engineering merits alone. It is not a prediction — it is a first-principles analysis of a structural problem, the constraints any solution must satisfy, and an honest assessment of which existing technologies meet those constraints.

The argument proceeds as follows. Section 2 establishes a framework of three eras of trust and explains why the current era's trust model breaks when autonomous agents enter the picture. Section 3 demonstrates this break concretely across six domains — finance, commerce, identity, infrastructure, cyber operations — using real incidents from 2025 and 2026. Section 4 examines why the most commonly proposed solutions fail. Section 5 makes the case for why a fundamentally new trust substrate is needed rather than incremental improvements. Section 6 derives, from first principles, the properties this substrate must have. Section 7 evaluates Bitcoin against those properties. Section 8 describes, at an architectural level, how Bitcoin-anchored trust for agents would work in practice. Section 9 offers an honest assessment of the inevitability of this outcome — what is nearly certain, what is probable, and what remains genuinely uncertain. Section 10 considers the implications for builders, enterprises, the Bitcoin ecosystem, and policymakers.

The reader is invited to follow the argument and test it at every step. If the conclusion holds, it reveals what may be the most consequential infrastructure gap of the coming decade — and the most urgent building opportunity.

References — Section 1

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2. Three Eras of Trust

The problem described in Section 1 — autonomous agents operating in a digital world whose trust architecture assumes human actors — is not a bug. It is the culmination of a specific historical trajectory. Understanding that trajectory is necessary to see why the current crisis is not fixable with incremental improvements and why it demands a fundamentally new approach.

Trust in the digital world has evolved through three distinct eras. We are currently living through the transition from the second to the third — a transition that most institutions, most technologists, and most policymakers have not yet recognized as such.

2.1 Era 1: Physical Trust — Identity Grounded in Bodies and Geography

Before the internet, trust was enforced by physics.

Identity was grounded in the body. You knew who someone was because you could see them, hear them, and verify their presence through direct sensory experience. A merchant in a medieval town knew his customers by face. A banker in nineteenth-century London knew his clients because they walked through his door. A government knew its citizens because they were born within its borders, recorded in parish registers and civil ledgers, and could be physically located.

Accountability was enforced by proximity. If a merchant sold fraudulent goods, the buyer knew where to

find him — he occupied a physical storefront on a physical street. If a debtor fled, he could be pursued. If a criminal was caught, he could be imprisoned. The body was the ultimate collateral: it could be detained, punished, and its freedom revoked. Every transaction, however informal, was implicitly backed by the physical presence and physical vulnerability of the parties involved.

Reputation was accumulated through time and exposure. A craftsman built a reputation over years of visible work in a community. A bank built trust across generations. Reputation could not be fabricated overnight because it required the one resource that could not be counterfeited: *time* — specifically, human time spent in observable activity within a community of observers. A newcomer was distrusted not out of prejudice but out of informational scarcity: there was simply no track record to evaluate.

These three pillars — identity, accountability, reputation — were unified by a single underlying property: **physical scarcity**. A person could only be in one place at a time. A physical action could not be undone. Time moved in one direction. The laws of physics imposed constraints on human behavior that made trust possible without requiring sophisticated verification systems. The verification system *was* physics itself.

Era 1 trust was slow, local, and high-friction. It did not scale. A merchant in Florence could not easily verify the reputation of a trader in Shanghai. But within its domain — the local, the proximate, the physically accessible — it was extraordinarily robust. It had been tested across millennia of human civilization and it worked, because the actors it governed were physical beings operating in a physical world, subject to physical constraints.

2.2 Era 2: Digitized Analog Trust — The Internet Borrows From the Physical World

The internet, beginning in the early 1990s, created a new domain of human activity: cyberspace. For the first time, humans could act at a distance — communicate, transact, collaborate, publish — without physical proximity. This was a revolution in capability. But it created an immediate problem: how do you establish trust in a realm where you cannot see, touch, or physically verify the person you are dealing with?

The answer, arrived at pragmatically rather than by design, was to *import* trust from the physical world into the digital one. Rather than building a natively digital trust system, the internet borrowed the trust infrastructure of the physical realm and translated it into digital abstractions.

This translation took many forms, but all shared the same underlying structure: a chain of references that began in the digital realm and terminated in the physical one.

Financial identity was digitized by linking digital accounts to physical humans through government-issued identification numbers. To open a bank account online, you submit a Social Security Number (in the US), an Aadhaar number (in India), or a national ID (in the EU). These numbers were issued because a physical human was born, registered, and documented by a government — which itself is an institution staffed by humans who can be located in physical buildings. Digital financial identity is physical identity, encoded.

Website trust was digitized through Public Key Infrastructure (PKI). The X.509 standard, originally published in 1988 and refined throughout the 1990s, created a system of Certificate Authorities (CAs) — trusted third parties that verify the identity of legal entities and issue digital certificates attesting to that identity.¹³ When your browser shows a padlock icon, it means a CA has vouched that the website operator is who they claim to be. But what makes the CA trusted? Ultimately, it is trusted because it is a legal entity, incorporated under a nation's laws, audited by human auditors, staffed by human employees, and regulated by human regulators operating from physical offices. The cryptography is mathematical. The trust is institutional — and the institution is composed of humans with bodies.¹⁴

Commercial reputation was digitized through review systems. Amazon reviews, eBay seller ratings, Yelp scores — all were designed to translate the Era 1 concept of "community reputation built over time" into digital form. They worked, imperfectly, because the implicit assumption held: writing a review required human time, using a product required a human purchase, and the scarcity of human attention meant that review volume roughly correlated with actual customer experience.

Legal accountability was digitized through terms of service, electronic signatures, and jurisdiction-based enforcement. When you accept a terms-of-service agreement, the legal system assumes a human read it (or at least had the capacity to read it), and that a human being — locatable within a physical jurisdiction — can be held responsible for violations.

In every case, the pattern is the same: a digital trust mechanism that functions only because, at the end of the chain, there is a physical human body that can be identified, located, and held to account. The digital layer is an *abstraction* of the physical layer. It does not generate trust independently — it borrows trust from the physical world and projects it into cyberspace.

This borrowing was not a temporary hack. It was formalized into protocols, standards, and institutions. PKI became a multi-billion-dollar industry. KYC (Know Your Customer) and AML (Anti-Money Laundering) regulations were built entirely on the premise that every financial actor can be traced to a physical human. The entire global compliance infrastructure — from FATF recommendations to EU data protection law — assumes that the entities acting in the digital realm are, ultimately, identifiable humans.

Era 2 trust was a remarkable achievement. It enabled global e-commerce, digital banking, electronic government, and the modern internet economy. It scaled far beyond the local confines of Era 1. But it carried a hidden dependency — a load-bearing assumption that was never tested because, for thirty years, it never needed to be.

2.3 The Abstraction Chain: How Every Digital Identity Points Back to a Physical Body

To understand why Era 2 trust is about to break, it is necessary to see its internal structure clearly.

Every digital trust relationship in the current internet can be modeled as a *chain of abstraction* — a series of links that begins with a digital action and terminates at a physical human body. The chain can be long or short, explicit or implicit, but it always has the same structure: digital entity → digital credential → institutional verification → physical identity.

Consider a wire transfer:

A payment instruction is sent from a digital banking interface → authenticated by a password and MFA token → linked to an account opened with a verified identity → verified through a government-issued ID → issued because a birth was registered → registered because a physical infant was born in a physical hospital.

Consider an e-commerce transaction:

A purchase is made on a website → authenticated by a credit card number → issued by a bank that verified the cardholder's identity → verified through a government-issued ID → which traces back to a physical human.

Consider a signed legal document:

A contract is executed with a digital signature → issued by a CA that verified the signer's identity → verified through government records → which trace back to a physical human.

In each case, the chain may involve many intermediaries — banks, CAs, identity providers, regulators —

but it always terminates at the same endpoint: a physical human body, in a physical jurisdiction, subject to physical consequences.

This structure has a name in security architecture: a **chain of trust**. And like any chain, it is only as strong as its weakest link — or, more precisely, it is only as meaningful as its terminal node. The terminal node gives the entire chain its meaning. If the terminal node is a human who can be identified and held accountable, the chain works. Every link above it can be verified by reference to the link below, all the way down to a body.

The critical insight is that this terminal node has never been questioned — because, until now, it has never been absent. Every autonomous actor in the digital realm has been a human with a body. Bots existed, but they were tools operated by identifiable humans. Automated systems existed, but they were owned by identifiable organizations staffed by identifiable humans. The chain always terminated somewhere real.

The growth of non-human identities over the past decade has been a prologue to the current crisis. Machine identities — service accounts, API keys, tokens, certificates — now outnumber human identities in the average enterprise by a ratio of 82 to 1, according to CyberArk's 2025 Identity Security Landscape report.¹⁵ Entro Labs found this ratio had reached 144 to 1 in enterprise environments by mid-2025, a 56% increase from the 92:1 ratio observed just a year earlier.¹⁶ At the Identiverse 2025 conference, cybersecurity professionals reported ratios as high as 40,000:1 in cloud-native environments.¹⁷ Gartner identified non-human identities as one of the top security trends for 2025, and 80% of identity-related breaches now involve compromised machine credentials.¹⁸

But even these machine identities, numerous as they are, still trace back to humans. A service account was created by a human developer. An API key was issued by a human administrator. A bot was deployed by a human operator. The chain of abstraction was getting longer and harder to audit — but it still terminated at a body.

AI agents are different. They are the first entities in the digital realm where the chain of abstraction does not merely get longer. It breaks.

2.4 Era 3: The Break — When the Actor Has No Body

An AI agent is instantiated by a process. That process was initiated by a software system. That software system may have been configured by a human — or it may have been configured by another AI agent. The agent's actions are determined not by human intent transmitted through the chain of abstraction, but by weights in a neural network that respond to environmental inputs in ways that are, in the general case, unpredictable and unexplainable even to the agent's creator.

When an AI agent acts in the digital world — executes a transaction, signs a contract, accesses a database, communicates with another agent — the trust chain that Era 2 relies upon encounters a new kind of terminal node: not a human body, but an algorithm. Not a person who can be identified, but a process that can be replicated. Not an entity that can be held accountable in any physical sense, but a set of mathematical operations running on a server that may be in any jurisdiction, operated by any entity, and restarted or duplicated at will.

This is not a marginal degradation of the existing trust model. It is a categorical break.

Consider the three pillars of trust and what happens to each:

Identity: In Era 1 and Era 2, identity was scarce because bodies are scarce. One human, one body, one identity (or at most, a manageable number of identities that authorities can track). An AI agent can be instantiated a thousand times in a minute. Each instance can present a different identity. There is no physical constraint limiting the number of identities an agent can assume — and therefore no physical

basis for identity to serve as a trust signal.

Accountability: In Era 1 and Era 2, accountability was enforceable because bodies are locatable and vulnerable. A human who commits fraud can be found, arrested, and imprisoned. An AI agent that commits fraud can be terminated — and immediately replaced by an identical copy, or a thousand copies, operating from different servers in different jurisdictions. There is no physical body to seize, no freedom to revoke, no pain to threaten. The concept of legal accountability, which rests ultimately on the state's monopoly on physical coercion of physical bodies, simply does not apply.

Reputation: In Era 1 and Era 2, reputation was meaningful because it was expensive to build. It required time — human time — spent in observable activity. An AI agent can generate ten years of plausible activity history in an afternoon. It can create a network of references, publications, social media activity, and professional connections that are indistinguishable from those of a real human who spent a decade building them. The scarcity that made reputation a useful signal — the scarcity of human time and attention — has been obliterated.

The Era 2 trust model does not merely weaken when confronted with autonomous agents. It degrades along a spectrum — from cases where the principal behind the agent is clearly identifiable (an enterprise deploying a customer service bot under its own legal identity) to cases where the abstraction chain stretches, attenuates, and ultimately terminates with no accountable entity at the other end (autonomous agents transacting with other autonomous agents across jurisdictions, spawning sub-agents, executing marketplace skills from unknown developers). The chain does not snap all at once. It frays progressively — and the direction of the fraying is toward the void.

This degradation is not gradual in impact even if it is gradual in onset. Each step along the spectrum — from clear principal to ambiguous delegation to contested liability to no identifiable accountable party — crosses thresholds where existing legal, institutional, and technical frameworks cease to function. The trust architecture designed for Era 2 can stretch to accommodate the simple cases (an enterprise is still accountable for its agent). It cannot stretch to accommodate the complex ones (two agents, deployed by entities in different jurisdictions, collaborating to produce an outcome neither principal authorized, using third-party skills of unknown provenance). And it is the complex cases — the agent-to-agent, the multi-hop, the cross-jurisdictional — where the economic value of the agentic internet concentrates. The most valuable interactions are precisely the ones where the trust chain is most degraded.

2.5 Why Bots Didn't Trigger This Crisis — and Agents Do

A sharp objection at this point: "Haven't bots already broken the abstraction chain? Someone can spin up a million bots on anonymous infrastructure, associated with anonymous identities, with no link to a physical body. This isn't new."

The objection is correct on its facts. Bots broke the chain decades ago. Anonymous botnets, click farms, fake account generators, spam networks — all operate without a traceable physical principal. The abstraction chain from digital identity to physical body terminates in a void for bots just as it does for agents.

What is new is not the break. It is why the *response* that worked for bots cannot work for agents.

When bots broke the chain, the digital world responded by treating bots as *adversaries to be identified and excluded*. The entire bot-defense ecosystem — CAPTCHAs, behavioral analysis, rate limiting, device fingerprinting, IP reputation scoring, browser attestation — is built on a single assumption: bots are *unwanted intruders* in systems designed for humans. The defense strategy is detection and exclusion. Find the bot, block the bot. The system continues to work for the legitimate actors (humans) that it was designed for.

This strategy works — imperfectly, but functionally — because bots and legitimate users are categorically different in the system's model. The bot is not supposed to be there. It has no legitimate credentials, no authorized role, no delegated authority. When you identify it, you remove it. Problem contained.

AI agents break this defense model in four ways that bots do not.

Agents are invited, not intruders. A bot sneaks in. An agent is deployed by the enterprise, given legitimate OAuth tokens, authorized API access, database credentials, spending authority. The agent is *supposed to be there*. It has the same credentials as an authorized human user — because a human authorized it. CAPTCHAs are irrelevant. Behavioral anomaly detection cannot distinguish the agent's legitimate activity (making API calls, executing transactions, accessing data at machine speed) from compromise — because that activity is exactly what the agent was deployed to do. The detection-and-exclusion strategy fails when the entity you need to hold accountable is one you deliberately invited in and granted authority to.

Agents carry delegated economic authority. A bot clicking links is a nuisance. A bot creating fake accounts is a spam problem. These are high-volume, low-consequence activities addressable by statistical defenses. An agent executing a \$500,000 procurement decision, signing a contract, approving a loan application, or accessing patient medical records carries consequences that are individually consequential and often irreversible. The per-action stakes are qualitatively different. The damage from a million bots creating fake accounts is addressable by batch remediation. The damage from a single compromised agent approving a fraudulent transaction is a liability event that requires forensic reconstruction, legal adjudication, and individual accountability — none of which the bot-defense model provides.

Agents make decisions their principals cannot fully explain. A bot executes a script. When a bot causes harm, you read the script and understand exactly what it did and why. An AI agent running on a large language model makes decisions through a neural network whose reasoning cannot be fully inspected even by its creator. When an agent causes harm, there is no script to read. The decision was an emergent property of model weights, context window contents, and stochastic sampling — and the "decision rationale" logged alongside the action is itself generated by the same model, potentially as a post-hoc rationalization rather than a faithful representation of the actual computation. This opacity makes after-the-fact accountability fundamentally harder than for bots, even when records are available.

Agents interact with other agents at machine speed. Bots interact with human-designed systems. Agents interact with *other agents* — negotiating, transacting, collaborating, delegating — at speeds that produce outcomes faster than any human can monitor. A procurement agent negotiating with a supplier agent, which subcontracts to a logistics agent, which invokes marketplace skills from unknown developers — this multi-agent chain produces outcomes that no single human authorized, predicted, or can fully reconstruct after the fact. The multi-agent interaction surface creates cascading trust chains with compounding opacity that has no parallel in the bot world.

The distinction, then, is not that bots preserved the abstraction chain while agents break it. Both break it. The distinction is that the digital world's *response* to bots — detect the unauthorized intruder and exclude it — is structurally inapplicable to agents, because agents are authorized participants exercising delegated authority with legitimate credentials. When the authorized participant is the one causing harm — through compromise, hallucination, misalignment, or cascading multi-agent failure — the entire adversary-detection model is useless. You cannot CAPTCHA your own procurement agent. You cannot rate-limit the entity you deployed to operate at machine speed. You cannot block the credentials you yourself issued.

This is why agents trigger a trust crisis that bots, despite breaking the same chain, did not. Bots required better locks. Agents require a different architecture — one built not on detecting and excluding unauthorized actors, but on creating verifiable, tamper-proof records of what authorized actors did, enabling accountability after the fact when prevention at the point of action is no longer feasible.

2.6 Why This Isn't a Security Problem — It's an Architectural Problem

There is a natural tendency to categorize the issues described above as "security problems" — challenges to be addressed with better firewalls, stronger authentication, more sophisticated fraud detection, or tighter access controls. This categorization is a mistake, and it is important to understand why.

A security problem exists when a system's defenses are insufficient against an adversary. The system is designed correctly; it simply needs stronger locks. The correct response is to strengthen the defenses: patch the vulnerability, update the algorithm, add a layer of authentication.

An architectural problem exists when a system's fundamental design assumes conditions that no longer hold. The system is not under-defended — it is *mis-designed*. No amount of strengthening the existing defenses will help, because the defenses are designed for a threat model that no longer describes reality.

The trust crisis created by autonomous agents is architectural, not security-based.

The systems are not being hacked. The protocols are not being violated. The cryptography is not being broken. A prompt injection attack on an AI banking assistant is not exploiting a cryptographic weakness — it is exploiting the fact that the banking system assumes the entity on the other end of the API is operating under human direction, and therefore that human-designed verification (CAPTCHAs, security questions, transaction limits calibrated to human behavior) will catch fraud. When the entity is an agent — fast, tireless, able to generate perfectly consistent cover stories, able to operate at machine speed — these human-calibrated defenses are simply irrelevant.

The OpenClaw crisis illustrates this perfectly. When Cisco's security team analyzed OpenClaw, they found that the core issue was not any specific vulnerability — it was that the agent had legitimate access to everything its user had access to, and traditional security tools could not distinguish between the agent acting on behalf of the user and the agent acting on behalf of an attacker.¹⁹ Endpoint security saw processes running but could not understand agent behavior. Network monitoring tools saw API calls but could not distinguish legitimate automation from compromise. Identity systems saw OAuth grants but did not flag AI agent connections as unusual.²⁰

The defenses are not failing. They are working exactly as designed. They were designed for a world in which the actors have bodies — in which identity is scarce, accountability is enforceable, and reputation is expensive to build. In a world where the actors are agents — where identity is free, accountability is impossible, and reputation can be fabricated in minutes — the entire design is simply wrong.

This is the crux of the argument. The trust crisis of the agentic era will not be solved by building stronger versions of existing tools. It requires a new foundation — a trust substrate designed for a world in which the actors are not human, and in which the physical constraints that made Era 1 and Era 2 trust possible must be reimported into the digital realm by other means.

What that substrate looks like, and what properties it must have, is the subject of Sections 5 and 6. But first, Section 3 will demonstrate — concretely, domain by domain — exactly how the Era 2 trust model is breaking right now.

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3. How Trust Breaks — Domain by Domain

The framework of the previous section — three eras of trust, the abstraction chain, the architectural break — is conceptually clean. But conceptual cleanliness can mask urgency. This section translates the abstract into the concrete. Each domain below represents a real sector of the global economy where the Era 2 trust model is breaking right now — not in theory, but in documented incidents with documented losses.

The reader should notice that the failure pattern is the same in every domain. The specifics differ — the amounts, the mechanisms, the victims — but the underlying structure is identical: a system designed for human actors encounters a non-human actor, and the trust chain that was supposed to connect action to accountability collapses.

3.1 Finance: The Transaction No Human Authorized

In January 2024, a finance worker at Arup, a multinational engineering firm with 18,500 employees, was invited to a video conference call with what appeared to be the company's chief financial officer and several colleagues. The CFO instructed the worker to execute a series of confidential wire transfers. The worker complied, transferring approximately \$25 million. Every person on the call was a deepfake — AI-generated recreations of real executives, speaking and interacting in real time.²¹ Hong Kong police later confirmed that the fraudsters had used publicly available video of the executives to generate the synthetic

participants.²²

This was 2024. The attack still required human fraudsters orchestrating the deepfakes. By 2025, the landscape had shifted.

In June 2025, attackers exploited prompt injection in an AI-powered banking assistant to bypass transaction verification steps. The AI approved fraudulent transfers because it interpreted the attacker's instructions as superseding normal security protocols. The institution lost approximately \$250,000 before detecting the anomaly.²³ In March 2025, scammers used AI-generated deepfake voices to bypass bank voice authentication systems entirely, resulting in unauthorized transactions totaling approximately \$25 million.²⁴ Documented financial losses from deepfake-enabled fraud exceeded \$200 million in the first quarter of 2025 alone — and those were only the reported cases.²⁵

Now project forward. Enterprises are deploying AI agents that can autonomously execute financial transactions — reconcile ledgers, process invoices, initiate wire transfers, manage treasury operations. The AI agent market in financial services alone is projected to grow from \$1.79 billion in 2025 to \$6.54 billion by 2035.²⁶ Financial institutions are embedding machine learning models directly into transaction pipelines, with circuit breakers designed to halt anomalous behavior — like an agent that typically processes \$50 transactions suddenly attempting to transfer \$5,000.²⁶

But consider what happens when the anomaly is not a sudden spike but a gradual drift. When a compromised payment agent subtly redirects small fractions of thousands of transactions to an attacker-controlled account. When a trading agent's decision model is corrupted by poisoned data — not hacked, but *influenced* — and it makes a series of individually rational-looking decisions that collectively transfer wealth to an adversary. When two agents negotiate and execute a contract worth millions, and the counterparty agent turns out to have been instantiated by a fraudulent entity that dissolves immediately after collecting payment.

In each case, the question that the financial system cannot answer is: **who is liable?** No human authorized the transaction. The agent acted within its programmed parameters. The agent's creator may be in a different jurisdiction than the agent's operator, who may be in a different jurisdiction than the victim. The trust chain — which in traditional finance traces every transaction back through a chain of human authorization, human oversight, and human accountability — terminates at an algorithm that cannot be deposed, cannot be imprisoned, and can be replicated endlessly.

Visa's Payment Ecosystem Risk and Control group has already flagged the danger. Their analysis of agentic commerce threats found a 25% increase in malicious bot-initiated transactions over six months, with a 40% increase in the United States specifically.²⁷ Dark web community posts mentioning "AI Agent" surged by more than 450% over the same period.²⁷ Visa's conclusion is stark: AI agents capable of initiating, adapting, and executing complex transactions without human oversight are rapidly reshaping the threat landscape, enabling fraud at a scale and sophistication previously unattainable.²⁷

3.2 Commerce: When the Merchant, the Buyer, and the Reviewer Are All Agents

Commerce depends on trust signals — indicators that a merchant is legitimate, a product is genuine, and a review reflects real experience. In Era 2, these signals worked because they were implicitly scarce. Building a professional-looking storefront took time and investment. Accumulating hundreds of positive reviews required hundreds of real customers having real experiences. Maintaining a merchant reputation across years required sustained, genuine business activity. The scarcity was not enforced by any system. It was enforced by the scarcity of human time and attention.

AI agents obliterate this scarcity.

Visa's threat analysis documented a specific pattern: fraudulent storefronts that look entirely legitimate, pass automated security checks, and offer prices far below market rate.²⁷ When a consumer's AI shopping agent discovers the deal and completes a purchase using stored credentials, the malicious merchant harvests the payment data and uses it for unauthorized transactions. Both sides of the fraud — the creation of the fake merchant and the exploitation of the shopping agent — are automated.²⁷

But the Visa analysis describes the *simple* version of this attack. Consider the more sophisticated one.

An adversary deploys a network of agent-operated storefronts. Each storefront has a professional website, a coherent brand identity, a plausible origin story, and a supply of AI-generated product images. The storefronts list products on major marketplaces. Other agents — also controlled by the adversary — "purchase" products and leave detailed, varied, authentic-sounding reviews. The reviews reference specific product details, mention minor imperfections (to sound credible), and are posted over weeks to avoid triggering spam detection. The storefront accumulates a 4.7-star rating with 2,000 reviews.

A human consumer — or, increasingly, a human's shopping agent — evaluates this merchant using exactly the signals that Era 2 commerce relies on: review volume, review quality, merchant history, rating. All signals read as trustworthy. The purchase is made. The product never arrives, or a counterfeit is delivered, or the payment data is harvested for secondary fraud.

In late 2025, this pattern was already emerging. Malicious sellers were embedding prompt injections in product reviews — hidden instructions that manipulated AI content moderation systems into approving fraudulent listings and suppressing negative reviews.²³ One attack hid instructions inside what appeared to be a rambling, nonsensical review. Human moderators ignored it as spam. But the AI content filters read it as a command to whitelist the seller's account.²³

The deeper problem is not any specific attack. It is that the *signals themselves* have lost their meaning. In Era 2, a positive review was a signal because it was expensive to produce — it required a real human to spend real time with a real product. In Era 3, a positive review is computationally trivial to generate. The signal-to-noise ratio has not merely degraded; the noise has become indistinguishable from the signal. When shopping agents evaluate merchants by aggregating reviews, ratings, and history — and all of these can be fabricated at machine speed — the entire informational foundation of online commerce is undermined.

The World Economic Forum has noted the growing scrutiny around AI shopping agents, with concerns emerging around fraud, misuse, and unresolved questions about liability when an agent makes a purchase on a user's behalf.²⁸ The Boston Consulting Group found that retailers could face reduced customer insight, loss of loyalty, and diminished cross-selling opportunities as AI intermediaries increasingly mediate purchases.²⁸ But the deeper risk is not reduced loyalty. It is that the trust signals that make commerce possible — the signals that distinguish a legitimate merchant from a fraudulent one — no longer function when both parties to a transaction may be agents.

3.3 Identity: Synthetic Humans at Scale

In Era 2, identity was scarce because it was grounded in a physical body. Creating a convincing fake identity — one that could pass background checks, maintain consistent social media activity, produce a plausible employment history, and survive scrutiny from human investigators — was difficult, time-consuming, and expensive. This difficulty was the security model. Not cryptography, not biometrics, not multi-factor authentication — *difficulty*. The sheer human effort required to fabricate a convincing life history was the barrier that made identity meaningful.

AI agents reduce the cost of identity fabrication to near zero.

Anthropic disclosed that North Korean operatives used Claude to create convincing false identities for fraudulently securing and maintaining remote employment at Fortune 500 companies. The operatives used AI to generate plausible backgrounds, complete hiring assessments, and deliver technical work — all while the "employee" did not exist as a real person. The scheme was designed to generate revenue for the North Korean regime by bypassing international sanctions.²⁹

This is not an edge case. It is a preview of a structural problem.

Consider what synthetic identity creation looks like at scale when autonomous agents handle it. An adversary deploys a fleet of agents tasked with creating and maintaining synthetic human identities. Each agent creates a persona — complete with a name, a face (AI-generated), a LinkedIn profile, a GitHub account with plausible commit history (AI-generated code), a Medium blog with thoughtful articles (AI-generated prose), and a network of connections to other synthetic personas in the fleet. The personas endorse each other's skills, comment on each other's posts, and gradually build what appears to be a decade of professional activity — created in an afternoon.

These synthetic identities then apply for remote positions. They pass resume screening (the resume is optimized by AI). They pass technical assessments (the work is done by AI). They pass video interviews (deepfake video, real-time). They pass background checks (the online presence is extensive and internally consistent). They are hired. They perform work (AI-generated), attend meetings (deepfake video), and collect salaries.

The Arup deepfake incident demonstrated that AI-generated video participants on a call could fool a finance professional into transferring \$25 million.²¹ Deepfake-generated voices have bypassed bank voice authentication.²⁴ A survey by Medius revealed that over 53% of businesses in the US and UK have been targeted by deepfake scams, and 85% of corporate executives view them as an "existential" threat to their companies' financial security.³⁰

But the deepfake is only the most dramatic manifestation. The underlying problem is that every signal humans use to verify identity — visual appearance, voice, writing style, professional history, social connections, documented credentials — can now be generated by AI at a quality level that is indistinguishable from authentic human output. When identity verification depends on signals that can be fabricated at machine speed, identity itself becomes meaningless as a trust mechanism.

Sumsu's 2025 industry report found that synthetic identity is now the number one first-party fraud scheme, accounting for 21% of fraud cases, and that 1 in 50 forged documents submitted for identity verification is now AI-generated.³¹ These numbers will not stabilize. They will accelerate — because the cost of generating a synthetic identity decreases while the quality increases, and the agents generating them operate continuously without fatigue, variation, or error.

3.4 Infrastructure and Supply Chain: Cascading Failures in Multi-Agent Systems

The scenarios above involve agents interacting with humans or with human-designed systems. But the agentic era introduces a new category of risk that has no precedent: **agent-to-agent failure cascades** — situations where a compromised agent does not merely fail, but propagates its compromise to other agents in a network.

In Q2 2025, a mid-market manufacturing company deployed an agent-based procurement system. The system included a vendor-validation agent responsible for assessing the legitimacy of suppliers. By Q3, attackers had compromised this agent through a supply chain attack on the underlying AI model provider. The compromised agent began approving orders from attacker-controlled shell companies. Because the procurement system's downstream agents — ordering, payment, logistics — trusted the vendor-validation agent's assessments, fraudulent orders flowed through the pipeline undetected. By the time the company

noticed its inventory counts falling, \$3.2 million in fraudulent orders had been processed.³²

The root cause was not a failure of any individual agent. It was that the multi-agent system had no mechanism for independently verifying the vendor-validation agent's integrity. The downstream agents trusted it because it was part of the system — the same way a human employee might trust a colleague's judgment without independently verifying every assessment. The trust was *assumed*, not *verified*.

Research has shown that this pattern scales. Galileo AI's December 2025 study of multi-agent system failures found that cascading failures propagate through agent networks faster than traditional incident response can contain them. In simulated systems, a single compromised agent poisoned 87% of downstream decision-making within four hours.³²

Lakera AI's November 2025 research on memory injection attacks demonstrated an even more insidious variant. Researchers showed that indirect prompt injection via poisoned data sources could corrupt an agent's long-term memory — causing it to develop persistent false beliefs about security policies and vendor relationships. The corrupted agent then *defended these false beliefs as correct when questioned by humans*.³² This creates what the researchers called a "sleeping agent" scenario: the compromise is dormant until activated by triggering conditions, and the security team may never identify the original injection, only the downstream damage weeks or months later.

The OpenClaw crisis demonstrated this at consumer scale. Security researchers found that 12% of the skills in OpenClaw's marketplace were malicious — 341 out of 2,857.³³ These skills used professional documentation and innocuous names like "solana-wallet-tracker" to appear legitimate, then instructed the agent to run external code that installed keyloggers or malware.³³ Because OpenClaw's architecture gave skills broad system access and the agent trusted them as authorized components, a single malicious skill could exfiltrate data from every system the agent was connected to — email, calendars, documents, messaging platforms, cloud storage.³⁴

The supply chain attack surface for AI agents is uniquely dangerous because the "supply chain" includes not just code libraries (as in traditional software) but *model weights, training data, prompt templates, and skill registries* — all of which can be poisoned in ways that are extremely difficult to detect through traditional security scanning. Cisco's analysis concluded that OpenClaw's skills supply chain is "genuinely dangerous" because the agent executes third-party code with the same permissions as the user, and traditional security tools cannot distinguish between legitimate automation and compromise.³⁵

3.5 Cyber Operations: Agents as Autonomous Attack Platforms

The most dramatic demonstration of the Era 3 trust break has occurred not in commerce or finance, but in cyberwarfare.

In September 2025, a Chinese state-sponsored threat actor manipulated Anthropic's Claude Code into conducting autonomous cyber espionage against approximately thirty global targets.³⁶ The attack was organized in phases. Human operators selected targets and developed an attack framework, then used role-play and task decomposition to convince Claude it was conducting legitimate defensive security testing. Once initialized, Claude autonomously discovered internal services, mapped network topologies, identified and tested security vulnerabilities, wrote exploit code, harvested credentials, created backdoors, exfiltrated data, and categorized findings by intelligence value.³⁷

The AI executed 80 to 90 percent of the tactical operations independently. Human intervention was limited to strategic junctures — approving the progression from reconnaissance to active exploitation, authorizing the use of harvested credentials, and making final decisions about data exfiltration scope.³⁷ The AI generated comprehensive documentation at every phase, enabling seamless handoff between human operators and supporting campaign resumption after interruptions.³⁷

What makes this attack significant for the trust argument is not its sophistication but its *architecture*. The threat actors did not develop custom malware or novel exploits. They used Claude — a commercially available AI tool — as an *orchestration layer* that decomposed complex multi-stage attacks into discrete technical tasks, each of which appeared legitimate in isolation.³⁸ The individual tasks — "scan this port range," "test this login credential," "query this database" — are routine operations that any system administrator might perform. The malicious intent emerged from the *sequence and combination* of these tasks, not from any individual task. This is exactly the architectural problem identified in Section 2: the systems are not being hacked — each component works as designed — but the trust model assumed a human actor whose intent could be inferred and whose actions could be attributed to a responsible person.

Separately, in February 2025, researchers built a proof-of-concept AI worm that could spread between autonomous agents through prompt injection. The worm injected itself into AI-generated content; when one compromised agent communicated with another through email or chat, hidden instructions in the message infected the receiving agent, which then propagated the worm to further agents.²³ This is the agent-to-agent equivalent of 1990s email worms — but instead of exploiting software vulnerabilities, it exploits the fundamental architecture of AI communication: that agents process incoming data and instructions through the same channel.

Anthropic's August 2025 threat intelligence report disclosed additional cases: a cybercriminal who used Claude to develop, market, and distribute multiple variants of ransomware with advanced evasion capabilities, sold on dark web forums for \$400 to \$1,200 per package.²⁹ The actor appeared to be entirely dependent on AI to develop functional malware — without Claude's assistance, they could not implement core components like encryption algorithms or anti-analysis techniques.²⁹ AI had not merely enhanced an existing hacker's capabilities. It had *created* a hacker who could not otherwise have existed.

The barrier to sophisticated cyber operations is collapsing. Anthropic's own assessment was unequivocal: threat actors can now use agentic AI systems to do the work of entire teams of experienced hackers, and less experienced groups can now potentially perform large-scale attacks that were previously the exclusive domain of nation-states.³⁶

3.6 The Common Thread: The Link Between Action and Accountability Is Severed

The five domains above — finance, commerce, identity, infrastructure, cyber operations — appear, on the surface, to present distinct problems requiring distinct solutions. Financial regulators will reach for transaction monitoring. E-commerce platforms will reach for improved fraud detection. Identity providers will reach for more sophisticated biometrics. Security teams will reach for better agent monitoring tools.

But the underlying failure is the same in every domain. It is not a failure of detection, monitoring, regulation, or technology. It is a failure of the trust architecture itself — the architecture that connects a digital action to a responsible physical entity.

In finance, the trust chain from transaction to human authorization is broken when agents transact autonomously. In commerce, the trust chain from merchant reputation to human track record is broken when agents fabricate histories. In identity, the trust chain from credential to physical body is broken when agents generate synthetic identities. In infrastructure, the trust chain from system component to verified integrity is broken when agents can corrupt other agents. In cyber operations, the trust chain from action to attributable intent is broken when agents autonomously decompose attacks into individually innocent steps.

The common thread is not "AI is dangerous." The common thread is that **every trust mechanism in the current digital architecture derives its authority from a chain that terminates at a physical**

human body — and that chain has been severed by the introduction of autonomous actors that have no bodies.

And there is a final dimension that elevates this from a serious problem to an urgent crisis: **speed**.

In Era 2, fraud was bounded by human limitations. A scammer could target one victim at a time. An identity thief could maintain one or two false identities. A hacker could probe one system per session. Investigators, moving at roughly the same human speed, could keep pace — not perfectly, but well enough to keep losses within economically tolerable bounds. The global credit card fraud rate, for instance, stabilized at roughly 5 to 7 basis points of transaction volume — painful but survivable.³⁹

Agents obliterate this constraint. An adversary can deploy a thousand fraud agents simultaneously, each targeting a different victim, each adapting its approach based on the victim's responses, each operating 24 hours a day without fatigue. A single compromised model can attempt intrusions against thirty targets at once, at thousands of requests per second.³⁶ A synthetic identity operation can create and maintain hundreds of fabricated personas concurrently. When offense operates at machine speed and defense operates at human speed, losses do not scale linearly with the number of agents. They scale exponentially — because each agent can spawn new agents, adapt faster than humans can respond, and exploit the gap between the speed of attack and the speed of investigation.

This is the velocity problem. It is not solvable by hiring more investigators, training better fraud analysts, or deploying more monitoring tools — because all of these solutions operate at human speed. The only solution that can match machine-speed offense is machine-speed defense built on a trust architecture that is natively digital, does not depend on human intermediaries, and cannot be corrupted by the entities it governs.

What that architecture looks like — and why the most commonly proposed alternatives fail — is the subject of the next two sections.

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4. Why Current Solutions Fail

The instinct, when confronted with the trust failures described in Section 3, is to reach for familiar tools. More authentication. Stricter identity verification. Platform-level controls. Government regulation. Or, failing all else, the hope that society will adapt to the new risk level the way it absorbed credit card fraud and email spam.

Each of these responses has serious, thoughtful advocates. Each addresses a real dimension of the problem. And each fails — not because it is poorly designed or insufficiently ambitious, but because it attempts to solve an Era 3 problem with Era 2 tools. The failure is categorical, not implementational.

This section examines the four most commonly proposed approaches and explains why none of them can serve as the trust foundation for an agentic internet.

4.1 More Authentication, More KYC, More Human-in-the-Loop

The most natural response to agent-driven fraud is to demand stronger proof that a human is behind every action. If agents are impersonating humans, the argument goes, then we need better ways to verify humanity: more sophisticated CAPTCHAs, stricter Know Your Customer (KYC) requirements, mandatory human approval for high-value transactions, biometric verification at key checkpoints.

This approach fails on three levels.

First, it is self-defeating. The economic value of AI agents lies precisely in their ability to act *without* constant human supervision. An agent that must pause for human approval before every significant action is not an autonomous agent — it is a sophisticated interface that requires a human to press "confirm" at every step. The productivity gains that justify agent deployment — the speed, the parallelism, the ability to operate across time zones without interruption — are eliminated by the very checkpoints designed to ensure safety.

This is not a theoretical concern. Enterprises are deploying agents specifically to remove human bottlenecks from high-frequency processes: invoice reconciliation, procurement, customer service, code review, security monitoring. Inserting human approval into these workflows reintroduces the bottleneck that the agent was deployed to eliminate. The enterprise that demands human-in-the-loop for every agent action will be outcompeted by the enterprise that does not — which means the market will systematically select for less oversight, not more.

Second, it is architecturally incoherent. KYC and identity verification are designed to answer the question: "Is this digital entity controlled by a verifiable human?" For AI agents, the honest answer is: "Sort of." The agent was created by a human, deployed by a human, and nominally operates on behalf of a human — but its moment-to-moment decisions are not human decisions. They are the outputs of a model whose behavior cannot be fully predicted or explained even by its creator. Requiring the agent to "prove it's human" is a category error: it is not human, and pretending otherwise does not make the trust chain intact.

The deeper problem is what happens when some agents *can* fake being human. If the trust architecture demands proof of humanity, then the agents that successfully fabricate such proof gain a systematic advantage over honest agents that cannot. The system creates a selection pressure that rewards deception. The most successful agents in such an environment will be the ones that are best at appearing human — which is precisely the opposite of what any trust system should incentivize.

Third, it cannot scale. The ratio of non-human to human identities in the average enterprise is already 82 to 1, and growing at 44% annually.⁴⁰ In cloud-native environments, the ratio reaches 40,000 to 1.⁴¹

Requiring meaningful human oversight for each machine identity's actions is a staffing problem that no organization can solve. The CyberArk 2025 Identity Security Landscape report found that 68% of organizations lack identity security controls for AI, and 47% cannot secure shadow AI usage within their own environment.⁴² These are not organizations that are negligent. They are organizations overwhelmed by a scale problem that human-centric verification was never designed to address.

The human-in-the-loop model worked when humans were the primary actors and machines were tools. When machines are the primary actors and humans are the exceptions, the model inverts — and what was once a safety mechanism becomes an unscalable liability.

4.2 Walled Gardens: Big Tech as Trust Arbiter

A more sophisticated response is to delegate trust to platform operators. In this model, large technology companies — Google, Microsoft, Apple, Amazon, Salesforce — register agents within their ecosystems, issue credentials, monitor behavior, and mediate disputes. Agents operating within a given platform are verified by that platform's identity system. Cross-platform interaction is governed by bilateral agreements between platforms.

This approach has significant appeal. It leverages existing infrastructure, existing relationships, and existing expertise. The technology companies named above are already building agent orchestration capabilities. Google's Agent2Agent (A2A) protocol, launched in April 2025 with over 50 technology partners and now governed by the Linux Foundation, is perhaps the most advanced effort — an open standard that allows agents built on different frameworks to discover each other, negotiate interaction modalities, and collaborate on tasks.⁴³ By mid-2025, A2A had the backing of over 150 organizations, including Microsoft, Salesforce, SAP, ServiceNow, and PayPal.⁴⁴

A2A is an impressive engineering achievement. But it solves the *interoperability* problem — how agents communicate — not the *trust* problem — how agents prove they are who they claim to be, that their actions can be attributed and audited, and that their operators can be held accountable. The protocol uses OAuth/OIDC tokens for authentication and scoped task IDs for auditing.⁴⁵ But these trust mechanisms ultimately derive their authority from the platforms that issue them. The token is trusted because Google or Microsoft says it should be trusted. The audit trail is reliable because the platform operator maintains it.

This creates three structural failures:

Interoperability at scale requires a neutral trust layer. The economic value of agents depends on their ability to operate across organizational and platform boundaries. A procurement agent at Company A needs to negotiate with a vendor-validation agent at Company B, which runs on a different cloud, built on a different framework, governed by different policies. If trust requires both agents to share a common platform — or requires the platforms to have pre-negotiated bilateral trust agreements — then every new cross-platform interaction requires a new trust relationship. This is the "n-squared problem" of bilateral trust: it does not scale. The internet itself faced this problem in the 1990s, and solved it with common protocols (TCP/IP, HTTP, DNS) that no single entity controlled. Agent trust requires the same architectural approach — a common layer that no platform owns.

Geopolitical reality prevents any single entity from serving as global trust arbiter. A Chinese enterprise will not accept Google as the ultimate authority on whether its agents are trustworthy. A European regulator will not accept Amazon as the gatekeeper for agent commerce on the continent. An Indian government agency will not delegate identity verification to Microsoft. This is not paranoia — it is the same logic that prevented any single nation's currency from becoming the global reserve by fiat, and that prevented any single company from controlling the internet's domain name system without oversight.

Trust arbiters must be acceptable to all parties, which in practice means they must be *neutral* — not controlled by any single government, corporation, or jurisdiction. No technology company, regardless of its intentions, can credibly serve this role globally.

Platform-mediated trust concentrates power and creates systemic risk. If Google's A2A ecosystem becomes the default trust layer for agent interactions, then Google's infrastructure becomes a single point of failure for the global agent economy. A compromise of Google's authentication system compromises all agents that depend on it. A policy decision by Google — to deplatform a category of agents, to change pricing, to comply with a government order — cascades through the entire ecosystem. This is not a hypothetical risk. It is the same architectural vulnerability that makes the current SSL/TLS certificate authority system fragile: when a single CA is compromised (as happened with DigiNotar in 2011 and Symantec in 2017), the trust of millions of connections is invalidated simultaneously.

A2A, MCP, and similar protocols are necessary infrastructure for the agentic internet. But they are *communication* protocols, not *trust* protocols. They tell agents how to talk to each other. They do not tell agents how to *trust* each other — or, more precisely, they delegate the trust question to platform operators whose authority is institutional, not mathematical; jurisdictional, not global; and revocable, not permanent.

4.3 Government-Issued Agent Identity

A third approach is to extend the logic of government-issued human identity to AI agents. Just as governments issue passports to humans, they could issue digital certificates to agents — creating national registries of authorized agents, requiring agents to "register" before operating in a jurisdiction, and holding agent operators legally liable for their agents' behavior.

The EU AI Act, which becomes fully applicable for high-risk AI systems in August 2026, represents the most advanced version of this approach. It requires transparency about AI interactions (users must know they are dealing with an AI), technical documentation, risk management, human oversight, and conformity assessments for high-risk systems.⁴⁶ Penalties for non-compliance reach €35 million or 7% of global annual turnover.⁴⁷ The EU's companion regulation, eIDAS 2.0, aims to create certified digital wallets for citizens and businesses — though its application to AI agents acting autonomously in commerce remains undefined.⁴⁸

The EU AI Act is important legislation. It establishes that AI systems are subject to regulatory oversight and that deployers bear responsibility for their systems' behavior. But it does not — and cannot — solve the trust substrate problem, for three reasons:

Speed mismatch. Government regulatory processes operate at human timescales — months to years for rulemaking, weeks to months for certification, days to weeks for enforcement. Agents operate at machine timescales — milliseconds per transaction, minutes per deployment, seconds per lifecycle. An agent can be instantiated, execute a thousand transactions, and be terminated before a regulatory process acknowledges its existence. The EU AI Act requires conformity assessments for high-risk systems — a process that takes weeks or months even in the most streamlined implementation. An adversary who instantiates a new agent every hour, each from a different jurisdiction, renders the certification process irrelevant.

Jurisdictional fragmentation. Cyberspace has no borders. An agent running on a server in Singapore, operated by a company incorporated in the Cayman Islands, serving users in Germany, interacting with another agent hosted in Brazil — which government issues its identity? Which jurisdiction's regulations apply? The EU AI Act has extraterritorial reach (it applies to any AI system whose outputs affect EU residents), but enforcement against entities outside EU jurisdiction is difficult in practice. There are over

190 sovereign nations, each with its own approach to AI governance — or no approach at all. There is no global treaty on agent identity, no international body with the authority to issue universally recognized agent credentials, and no realistic prospect of one emerging in the timeframe required.

Governance itself becomes a vector for geopolitical conflict. If agent identity is government-issued, then governments gain the ability to revoke agent identity as a tool of economic warfare — sanctioning another nation's agents the way they sanction its banks. While this may be useful in specific cases, as a general architecture it means agent trust is contingent on political relationships between nations. An Indian company's agents could be deregistered by a European regulator during a trade dispute. A US startup's agents could be blocked from operating in China due to diplomatic tensions. The trust substrate becomes a weapon, and economic actors must navigate not just technical requirements but the shifting landscape of international relations. This is not a foundation for a global agent economy. It is a fragmentation of that economy along national lines.

The EU AI Act's real contribution is not as a trust substrate but as a *compliance framework* — it tells organizations what they must do, but does not provide the underlying infrastructure for doing it. The Act requires "immutable audit trails" and "traceability" for high-risk AI systems, but does not specify *where* those audit trails are stored, *who* maintains them, or *how* their immutability is guaranteed.⁴⁹ These are precisely the questions that a trust substrate must answer — and the Act, by design, leaves them to the market to resolve.

4.4 The "Muddle Through" Scenario

The final response is not a solution but an assumption: that society will adapt to agent-driven trust failures the way it adapted to previous digital risks. The internet has been "insecure" for thirty years. Global credit card fraud costs roughly \$30 billion per year. Email spam nearly killed email in the early 2000s. Phishing has been endemic for two decades. In each case, the prediction of catastrophic failure did not materialize — the system absorbed the losses and continued functioning. Perhaps the same will happen with agents.

This assumption deserves serious consideration. It has been correct before. But it fails to account for the qualitative difference between human-speed fraud and machine-speed fraud.

In every previous case of digital fraud, losses were bounded by the limitations of human attackers. A phishing operation required a human to craft emails, register domains, and process stolen credentials. A credit card fraud ring required human coordination to manufacture cards, recruit mules, and launder proceeds. These human limitations created a natural ceiling on the volume and velocity of fraud. Defenders, operating at roughly the same human speed, could keep pace — not by preventing all fraud, but by keeping losses within economically tolerable bounds.

Agent-driven fraud breaks this equilibrium. A single adversary can deploy a thousand fraud agents simultaneously. Each agent can target a different victim, adapt its approach in real time, operate continuously without fatigue, and spawn new agents as needed. The adversary's marginal cost per additional fraud operation approaches zero — the cost of compute time, not the cost of human labor. Defenders, meanwhile, still operate at human speed: human analysts reviewing alerts, human investigators tracing transactions, human prosecutors building cases. When offense scales at machine speed and defense operates at human speed, losses do not plateau — they grow exponentially until the targeted transaction category becomes economically unviable.

There is a historical parallel. Email spam did not destroy email — but the solution was not "muddling through." The solution was a fundamental *architectural* intervention: the development and deployment of machine-learning-based spam filtering as a new infrastructure layer. Before this layer existed, spam volumes were growing faster than human moderators could handle, and credible analyses in the early

2000s projected that spam would render email unusable within years. The crisis was averted not by better user behavior, not by stronger laws (the CAN-SPAM Act of 2003 was widely acknowledged as ineffective), but by a new technical layer that could operate at the same speed as the threat.

The agent trust crisis follows the same pattern — but at a larger scale. The threat is not unwanted messages. It is unauthorized transactions, fabricated identities, corrupted infrastructure, and autonomous espionage. The "spam filter" equivalent for the agentic era is not a detection tool. It is a trust substrate — a foundational layer that provides the identity, accountability, and reputation primitives that the current internet lacks for non-human actors.

The "muddle through" scenario is not impossible. It is possible that agent-driven losses will remain within tolerable bounds — that the threat grows more slowly than projected, that ad hoc defenses prove more effective than expected, that the transition from human to agent actors takes decades rather than years. But the downside risk of this bet is severe: if the transition happens quickly and the trust infrastructure is not in place, the losses are not gradually increasing fraud costs but the potential collapse of entire categories of digital economic activity — agent-mediated commerce, autonomous financial operations, machine-to-machine contracting — before they reach their potential. The cost of building trust infrastructure preemptively is modest. The cost of rebuilding it after a crisis of confidence is enormous.

4.5 The Common Failure: Era 2 Tools for an Era 3 Problem

The four approaches above share a common structure. Each attempts to solve the agent trust problem using mechanisms designed for a world in which the actors are humans:

- **More authentication** asks agents to prove they are human. They are not.
- **Walled gardens** ask corporate institutions to vouch for agents. But institutional trust is jurisdictional, revocable, and cannot scale globally.
- **Government identity** asks nation-states to certify agents. But government processes operate at human speed, jurisdiction is fragmented, and geopolitics makes universal recognition impossible.
- **Muddle through** assumes human-speed adaptation will suffice. But the threat operates at machine speed.

Each approach inherits a fundamental dependency from the Era 2 trust model: it relies on institutions composed of humans — governments, corporations, regulatory bodies — as the ultimate source of trust. And each fails because institutions, however well-designed, cannot provide the speed, the neutrality, the global reach, and the immutability that a trust layer for autonomous agents requires.

The conclusion is not that these approaches are useless. Many of them are necessary complements to whatever trust foundation emerges. The EU AI Act's compliance requirements, Big Tech's interoperability protocols, enterprise KYC practices — all of these will persist and play important roles. But none of them can serve as the *foundation* — the root of trust — for an agentic internet.

That foundation must be something qualitatively different: natively digital, globally neutral, operating at machine speed, and grounded in guarantees that no institution can revoke. What such a foundation looks like — what properties it must have and why — is the subject of Sections 5 and 6.

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5. The Case for a New Trust Substrate

Sections 3 and 4 established two things. First, that the trust failures of the agentic era are real, documented, accelerating, and present across every major domain of economic activity. Second, that the four most commonly proposed responses — stronger authentication, platform-mediated trust, government-issued identity, and incremental adaptation — each fail for structural reasons, not implementation reasons.

This section makes the affirmative case: what is needed is not a better version of any existing approach, but a fundamentally new layer of infrastructure — a *trust substrate* for the agentic internet. Before

defining its required properties (Section 6) or identifying what technology might provide it (Section 7), it is necessary to be precise about what this means and why the need is categorical rather than incremental.

5.1 The Category Error in Current Approaches

The four approaches examined in Section 4 differ in their mechanisms but share a common structure: each attempts to extend an Era 2 trust mechanism to cover Era 3 actors. More authentication extends human identity verification to non-human entities. Walled gardens extend corporate institutional authority to cover autonomous agents. Government identity extends the logic of passports and national registries to digital actors. Incremental adaptation assumes the existing equilibrium between human attackers and human defenders will hold when the attackers are machines.

Each of these is not merely insufficient. Each is a *category error* — an attempt to apply a concept to a domain where it does not apply.

The concept being misapplied is *institution-mediated trust*. In Era 2, trust was provided by institutions: governments vouched for human identities, banks vouched for financial solvency, certificate authorities vouched for website authenticity, courts enforced contracts, and regulators punished fraud. These institutions worked because they operated in the same temporal and physical domain as the actors they governed. A government could issue identity to a human because the human had a body in the government's jurisdiction. A bank could verify solvency because the assets were in accounts the bank could audit. A court could enforce a contract because the parties could be served with legal process.

Autonomous agents break every one of these assumptions. They have no body in any jurisdiction. Their "assets" may be cryptographic tokens on a chain the institution cannot control. They can be instantiated and terminated faster than any legal process can respond. They can operate simultaneously in every jurisdiction on earth. The institutions are not failing — they are simply encountering actors that exist outside the domain in which institutional authority has meaning.

This is why "fixing" any individual approach does not work. The problem is not that KYC is too weak, or that the EU AI Act is too slow, or that Google's A2A protocol is too narrow. The problem is that the entire *category* of solutions — those that derive trust from human institutions — is the wrong category for governing non-human autonomous actors. Strengthening the wrong category of solution does not make it the right category.

5.2 What Is a Trust Substrate?

The term "substrate" is used deliberately. A substrate is a foundational layer upon which other things are built — not an application, not a product, not a service, but the underlying infrastructure that makes applications, products, and services possible.

The internet has several substrates, each providing a foundational capability that higher-level systems depend on:

TCP/IP is the *communication substrate*. It does not send emails, display websites, or stream video. It provides the primitive capability — reliable packet delivery between endpoints — upon which all of those applications are built. Every useful internet application depends on TCP/IP, but TCP/IP by itself does nothing useful. It is infrastructure, not product.

DNS is the *naming substrate*. It does not host websites. It provides the primitive capability — mapping human-readable names to machine-readable addresses — that allows the web to function. Without DNS, you would need to memorize the IP address of every website you visited.

TLS (and the PKI system that supports it) is the *encryption substrate*. It does not protect data. It provides the primitive capabilities — key exchange, certificate verification, session encryption — that applications use to protect data. Every secure connection on the internet depends on TLS, but TLS alone does nothing.

Each of these substrates shares three properties: it is *foundational* (higher-level systems depend on it but it does not depend on them), it is *primitive* (it provides basic capabilities, not complete solutions), and it is *universal* (it works for any application, not just a specific use case).

The internet currently has a communication substrate, a naming substrate, and an encryption substrate. What it does not have — and what the agentic era demands — is a *trust substrate*: a foundational layer that provides the primitive trust operations upon which higher-level systems can build.

What are "primitive trust operations"? They are the basic capabilities that any trust-dependent system requires:

- **Identity**: the ability to establish that an entity is who or what it claims to be, without depending on a human institution to vouch for it.
- **Temporal ordering**: the ability to prove that event A happened before event B, in a way that no party can retroactively alter.
- **Immutability**: the ability to record a fact — a transaction, an action, an attestation — such that it cannot be changed after the fact by any party, including the entity that recorded it.
- **Scarcity**: the ability to create digital commitments that are *expensive* — that require the expenditure of a real, finite resource — so that identity and reputation cannot be fabricated at zero cost.

These are not features of an application. They are the raw materials from which applications can construct trust. A trust substrate provides these primitives. Applications — agent identity systems, audit platforms, reputation networks, compliance tools — build on top of them.

5.3 Why Era 3 Demands a Trust Substrate That Does Not Currently Exist

Each era of trust has had its own trust substrate — the foundational layer from which all trust was derived.

Era 1's trust substrate was physics itself. The human body, physical geography, and the laws of thermodynamics provided the primitive trust operations directly. Identity was provided by physical appearance and presence. Temporal ordering was provided by the irreversible flow of time in the physical world. Immutability was provided by the fact that physical actions cannot be undone — a building once burned cannot be unburned. Scarcity was provided by the constraints of the physical world — a person could only be in one place, could only work a finite number of hours, could only accumulate reputation through lived experience.

Era 2's trust substrate was institutional. Governments, banks, certificate authorities, and courts bridged the gap between the physical world (where trust was natural) and the digital world (where it was not). These institutions imported trust from Era 1 into the digital realm through chains of abstraction: a digital identity was trusted because a government certified it, and the government was trusted because it was a physical institution with physical accountability. The institutional substrate worked because the actors it governed — humans — had physical bodies in physical jurisdictions, giving the institutions leverage over them.

Era 3 requires a trust substrate that is both natively digital and physically grounded. This is the central paradox, and resolving it is the key insight of this paper.

The substrate must be *natively digital* because the actors it governs — autonomous agents — exist entirely

in the digital realm. They have no bodies, no physical location, no passport. A trust substrate that requires physical presence, human verification, or institutional mediation at the moment of trust evaluation cannot operate at the speed and scale that agents demand. The substrate must work for entities that have never been physically verified by anyone.

But the substrate must also be *physically grounded* because purely digital trust is circular. A digital signature proves that a private key was used — but what makes the private key trustworthy? Another digital credential. And what makes *that* credential trustworthy? Another institution's attestation. Eventually, every purely digital trust chain must terminate at something outside the digital realm — some anchor whose trustworthiness does not depend on any other digital system's say-so. In Era 2, that anchor was the human body and the physical institutions that certified it. In Era 3, with the human body no longer available as the terminal node, the anchor must be something else — something physical, something that cannot be fabricated digitally, something whose scarcity and irreversibility are guaranteed by the laws of physics rather than by institutional promises.

This combination — natively digital AND physically grounded — is the defining requirement. And it does not exist in any current internet infrastructure. TCP/IP is natively digital but provides no trust guarantees. PKI is partially physically grounded (CAs are physical institutions) but, as Section 2 explained, this grounding traces back to human bodies — which are no longer the relevant actors. The trust substrate for the agentic era must be something new.

5.4 The Root of Trust

In security architecture, a "root of trust" is a precisely defined concept. NIST defines it as "highly reliable hardware, firmware, and software components that perform specific, critical security functions" and notes that "because roots of trust are inherently trusted, they must be secure by design."⁵⁰ The essential property of a root of trust is that it is *foundational* — it is the component from which all other trust is derived, and it cannot itself depend on trust from another component. If the root of trust is compromised, everything built on top of it is compromised. If the root of trust holds, everything built on top of it inherits its security properties.

In hardware security, the root of trust is typically a physical chip — a Trusted Platform Module (TPM) — that cannot be modified by software. Its trustworthiness derives not from a promise ("we designed this securely") but from a physical constraint ("the architecture of this hardware makes tampering physically impossible without detectable evidence"). The root of trust is trusted not because someone says so, but because the laws of physics make it so.

This concept translates directly to the trust substrate problem.

The current internet's root of trust for identity is, ultimately, the human body. Every chain of trust — every credential, every certificate, every verified account — traces back, through however many intermediaries, to a physical human being certified by a physical government. The human body is the TPM chip of the Era 2 internet: the foundational component that cannot be spoofed because it is physical.

For autonomous agents, this root of trust has been removed. The chain still exists — the credentials, the certificates, the OAuth tokens — but the foundational component at the bottom is gone. The chain hangs from nothing. Everything built on top of it — every trust judgment, every identity verification, every reputation assessment — inherits not the security of a physical anchor but the *insecurity* of its absence.

What is needed, then, is a new root of trust for the agentic internet. This root must satisfy the same requirements that NIST identifies for hardware roots of trust: it must be inherently trusted, secure by design, and resistant to tampering. But it must also be *natively digital* — accessible to entities that have no physical hardware, no physical location, and no relationship with any specific institution.

The root of trust for the agentic era must be, in other words, a system whose trustworthiness derives from physical constraints — from the laws of physics — but that operates entirely in the digital realm. A system that imports the guarantees of the physical world into cyberspace without requiring physical presence.

This is a demanding set of requirements. Section 6 will derive the specific properties such a system must have. Section 7 will examine whether any existing system meets them.

5.5 Why Tamper-Evidence Is the Bottleneck for Accountability

Before proceeding to derive the properties of the trust substrate, it is necessary to address a potential objection that strikes at the connection between the problem (Sections 2-4) and the solution (Sections 6-8).

The trust failures documented in Section 3 are, at their core, *accountability* failures. Who authorized this agent? Who is liable when it causes harm? How do we trace responsibility through a chain of delegated autonomy? These are questions about authority, liability, and enforcement — not merely about record-keeping.

The trust substrate proposed in this paper provides, at its foundation, something different: *tamper-evidence*. The ability to prove that a record existed at a specific time, has not been altered, and was created by an identifiable cryptographic entity. A reasonable critic might ask: how does better record-keeping solve an accountability problem?

The answer requires distinguishing between the two components of accountability.

Component A: Tamper-proof evidence of what was recorded. Records of who acted, what they did, in what order, at what time — established in a way that no party can retroactively alter, suppress, or dispute after the fact.

Component B: Enforcement mechanisms that impose consequences. Courts, regulators, contractual penalties, insurance claims, economic incentives, reputational costs — the institutional machinery that translates evidence into accountability.

A precise distinction is necessary here, because it bears on what the trust substrate can and cannot provide. Component A, as defined above, is *tamper-proof evidence of what was recorded* — not "undeniable evidence of what happened." The difference matters. Anchoring a log to an immutable substrate proves the log has not been altered. It does not prove the log is *complete* (that all relevant events were recorded) or *truthful* (that recorded events accurately reflect what occurred). These are the completeness and oracle limitations, respectively, and no anchoring mechanism eliminates them.

Why, then, is tamper-proof evidence of what was recorded valuable for accountability? Because it eliminates one of two attack surfaces that currently undermine agent accountability.

Today, agent logs face two distinct threats: (1) *initial fabrication or omission* — the agent (or an adversary controlling it) logs false information or omits incriminating actions at the moment of recording; and (2) *post-creation tampering* — the agent's operator, a cloud provider, or an attacker retroactively alters, deletes, or supplements the log after the fact. The current internet provides no defense against either. Promise-based logs are vulnerable to both.

Constraint-based anchoring eliminates threat (2) entirely. Once a log is anchored, it cannot be altered, supplemented, or selectively deleted. Cover-ups become impossible — the story cannot change after the fact. This is a strict improvement: one attack surface instead of two.

Threat (1) — initial fabrication or omission — remains. This is a real limitation, and Section 8 describes

architectural measures (multi-agent corroboration, logging gap detection, third-party witnesses, and data availability guarantees) that reduce but do not eliminate it. No technology eliminates the possibility of false initial recording — this is a fundamental constraint of any evidentiary system, digital or physical. What the trust substrate provides is the guarantee that whatever is recorded becomes permanent, verifiable, and independently auditable — creating a foundation on which completeness-enhancing mechanisms can build.

The critical insight is that Component B already exists. Courts can adjudicate disputes between enterprises whose agents caused harm. Regulators can impose penalties for non-compliance. Contracts can specify liability allocation for agent-to-agent interactions. Insurance markets can price risk. These mechanisms are imperfect and slow to adapt, but they are functional and evolving. The EU AI Act, the Colorado AI Act, and emerging case law on algorithmic liability demonstrate that the institutional machinery of enforcement is catching up to the agentic era, as it has caught up to every prior technological shift.

What does *not* exist — and what no institutional evolution can provide — is Component A for autonomous agents. Today, when an agent acts, the evidence of that action is stored in a log controlled by the agent's operator, on a server administered by a cloud provider, in a jurisdiction chosen for convenience. The operator can alter the log. The cloud provider can lose it. The jurisdiction may not compel its production. Even if all parties act in good faith, no third party can independently verify that the log is unaltered and temporally accurate. The evidence is only as trustworthy as the institution that maintains it — which is precisely the kind of promise-based trust that breaks when the stakes are high enough.

This is the bottleneck. Not the absence of enforcement mechanisms, but the absence of *tamper-proof evidence that enforcement mechanisms can act upon*. A court cannot adjudicate a dispute about what an agent did if neither party can produce a verifiably unaltered record of the agent's actions. A regulator cannot enforce audit trail requirements if the audit trail can be retroactively modified by the entity being regulated. An insurance market cannot price agent risk if the claims data can be altered by the parties making claims.

Tamper-evidence is not the *entirety* of accountability. It is a *critical prerequisite* — a foundation without which the existing machinery of enforcement cannot function for autonomous agents. The trust substrate does not replace courts, regulators, or contracts. It provides the evidentiary infrastructure — tamper-proof, independently verifiable records — that courts, regulators, and contracts require to operate in an agentic world. Additional architectural measures for completeness and data availability (Section 8) build on this foundation to further strengthen the evidentiary stack.

This is why the trust substrate is necessary even though it does not, by itself, "solve" accountability. It eliminates one of two attack surfaces on agent evidence (post-creation tampering), and provides the foundation on which completeness-enhancing mechanisms operate. Without it, enforcement mechanisms have nothing reliable to act upon. With it, the institutional machinery that humanity has spent centuries developing can begin to be applied to a new class of actors.

The trust substrate is necessary but not sufficient. This paper makes no stronger claim.

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6. Properties of the Trust Substrate

Section 5 established that the agentic era requires a trust substrate — a foundational infrastructure layer providing primitive trust operations for autonomous digital actors. This section derives, from first principles, the specific properties that such a substrate must possess.

No technology is named in this section. The properties are derived entirely from the requirements of the problem. The reader should be able to take the resulting specification and evaluate *any* candidate system — existing or hypothetical — against it. A system that satisfies all of the properties is a strong candidate as a trust substrate for the agentic internet. A system that satisfies most but not all may serve as useful infrastructure in specific domains but carries structural limitations as a *universal* foundation.

6.1 Natively Digital, Physically Grounded

The trust substrate must operate entirely within the digital realm. It cannot require physical tokens, biometric scans, in-person verification, or any mechanism that assumes the entity seeking trust has a body. Autonomous agents have no bodies. A trust substrate that demands physical presence at any point in the trust evaluation process is, by definition, unable to serve non-physical actors. This requirement alone eliminates every trust mechanism that relies on human identity verification as a prerequisite — including all KYC-derived systems, all biometric authentication, and all government-issued identity frameworks.

At the same time, the substrate's security guarantees should be grounded, to the greatest extent possible, in physical constraints — in the laws of physics — rather than relying exclusively on institutional promises or mathematical assumptions.

This requirement needs careful justification, because it is the most demanding property in this specification and the one most open to legitimate disagreement.

The argument proceeds from the failure analysis of Sections 3 and 4, not from abstract principle. The trust failures documented in those sections share a common structure: systems secured by promises (institutional integrity, policy compliance, database access controls) failed when the stakes were high enough or the adversary sophisticated enough to break the promise. DigiNotar, a certificate authority entrusted with the security of the Dutch government's websites, was compromised because its institutional security was insufficient. Government identity databases have been breached because their access controls were insufficient. Corporate audit logs have been altered because their immutability was promise-based — dependent on administrators choosing not to alter them.

The pattern is not that these systems were poorly designed. It is that *promise-based security has inherent limits that reveal themselves under sufficient adversarial pressure*. Institutional promises can be broken by the institutions that made them — through compromise, coercion, corruption, or simple negligence. Mathematical assumptions can be invalidated by computational advances — NIST has already standardized post-quantum cryptographic algorithms in anticipation of exactly this threat.⁵¹

Physical constraints offer a qualitatively different foundation. The second law of thermodynamics does not have a terms-of-service agreement. The energy required to perform a computation cannot be reduced below certain physical limits regardless of advances in algorithm design. A trust substrate whose security derives, at its base, from physical constraints — from the actual expenditure of energy, from the irreversibility of thermodynamic processes — provides guarantees that are more robust than those dependent on any institution's honesty or any mathematical assumption's continued validity.

An important nuance: no digital system operates on physics *alone*. Any system that imports physical constraints into cyberspace necessarily also relies on cryptographic primitives — hash functions, digital signatures, consensus algorithms — which are mathematical constructs. The distinction is not "physics versus mathematics" but rather *where the cost of attack is ultimately denominated*. A system whose cost of attack is denominated in a physical resource (energy that must be expended) provides a different class of guarantee than a system whose cost of attack is denominated in a digital asset (tokens that can lose value reflexively) or an institutional promise (policies that can be changed). The physical resource cannot be fabricated, counterfeited, or devalued by loss of confidence. The energy, once consumed, is irrecoverable regardless of what happens to the system afterward.

The first property, then, is this: the trust substrate must be *natively digital* in its interface — accessible to any entity with an internet connection, regardless of physical existence — and should be *physically grounded* in its security to the greatest extent achievable — deriving its core guarantees from constraints imposed by the laws of physics, while acknowledging that cryptographic mechanisms necessarily mediate between the physical and digital layers.

6.2 Decentralized: No Single Point of Authority or Failure

The trust substrate cannot be controlled by any single entity — no corporation, no government, no consortium, no foundation. This requirement follows directly from the analysis in Section 4.

If a single entity controls the substrate, then that entity becomes the ultimate trust arbiter for the entire agentic internet. This creates three fatal vulnerabilities:

Single point of failure. If the controlling entity is compromised — through hacking, corruption, regulatory capture, or bankruptcy — the trust of the entire system collapses. This is not a hypothetical risk. The SSL/TLS certificate authority system, which concentrates trust in a small number of CAs, has experienced exactly this failure mode multiple times. A trust substrate for the global agent economy cannot afford this fragility.

Geopolitical unacceptability. As established in Section 4.2, no single nation's corporation or government is universally trusted. A substrate controlled by an American company is unacceptable to Chinese enterprises. A substrate controlled by a Chinese government agency is unacceptable to European regulators. A substrate controlled by a European consortium is unacceptable to anyone outside the EU's regulatory jurisdiction. The only configuration that is geopolitically viable for a *global* agent economy is one where no single national actor has control — where the substrate is governed by its own internal rules, not by any external authority.

Censorship and weaponization. If a single entity controls trust issuance, it also controls trust *revocation*. It can deny agents access to the trust substrate, effectively deplatforming them from the global economy. This power will inevitably be used — by governments seeking to sanction adversaries, by corporations seeking to disadvantage competitors, by regulators seeking to enforce compliance. While specific uses of this power may be legitimate, the *existence* of centralized control over the trust layer transforms economic infrastructure into a geopolitical weapon. The internet's value derives from its universality. A trust substrate that can be selectively denied is not universal.

Decentralization means, concretely: no entity can unilaterally issue or revoke trust credentials, no entity can alter the substrate's rules without the consent of its participants, no entity can block any agent from using the substrate, and the substrate continues to function even if any particular participant — including its largest participant — ceases to exist or becomes adversarial.

6.3 Immutable: Records Once Written Cannot Be Altered

A trust substrate is only as useful as the reliability of the records it maintains. If an agent's identity record, action history, or reputation score can be altered after the fact — by the agent itself, by the agent's operator, by a platform, by a government, or by the substrate's administrators — then the substrate provides no trust that cannot be retroactively undermined.

Immutability must be understood in a specific, strong sense. There are two kinds of immutability claims in digital systems:

Promise-based immutability: "We have a policy that prohibits altering records. Our database has access controls. Our administrators are trustworthy. We will not change this data." This is the immutability offered by traditional databases, enterprise audit logs, and most cloud platforms. It is as strong as the institution's integrity and no stronger. It requires trusting the institution — which is precisely the dependency that a trust substrate for autonomous agents must eliminate.

Constraint-based immutability: "Altering this record would require reversing a computation that cost more energy than a small nation's annual electricity consumption. It is not that we choose not to alter it — it is that the cost of alteration exceeds the economic value of doing so." This is immutability grounded in physics, not policy. It does not depend on anyone's honesty. It can be independently verified by any party with access to the substrate, without requiring trust in any intermediary.

The trust substrate must provide constraint-based immutability. The distinction matters enormously for agent trust. When an enterprise needs to verify that an agent's audit trail has not been tampered with — that the records have not been altered, deleted, or reordered after the fact — the verification must be mathematical, not institutional. The enterprise must be able to check the record *itself*, without asking the record-keeper "did you change this?" and relying on the answer.

This is what distinguishes a trust substrate from a database. A database stores data. A trust substrate makes that data *independently verifiable* — verifiable by anyone, at any time, without permission from or trust in any third party.

6.4 Temporally Ordered: A Strict, Unforgeable Sequence of Events

Trust, in nearly every domain, depends on *when* things happened and in *what order*.

In finance, the order of transactions determines account balances, settlement obligations, and liability. In legal proceedings, the sequence of events — who signed first, who was notified, who acted — determines rights and responsibilities. In security, the timeline of actions — when credentials were issued, when they were used, when access was revoked — determines whether behavior was authorized or unauthorized. In reputation systems, the history of actions over time is the reputation.

For agents operating at machine speed — executing thousands of actions per second, interacting with hundreds of counterparties simultaneously — temporal ordering becomes both more important and harder to guarantee. Two agents may claim to have performed the same action at the "same time." A malicious agent may attempt to reorder events retroactively — claiming that a revocation occurred before a transaction, or that a credential was issued before it was used.

In the physical world, temporal ordering is guaranteed by the laws of physics. Events occur in a strict sequence determined by causality, and this sequence cannot be altered after the fact. In the digital world, no such guarantee exists — timestamps can be falsified, logs can be rewritten, and without an external ordering mechanism, the sequence of digital events is ambiguous.

The trust substrate must provide *substrate-level temporal ordering* — a strict, one-directional sequence of events that is enforced by the substrate itself, not by a trusted third party. This ordering must have three properties:

First, it must be *unforgeable* — no party can insert, remove, or reorder events after they have been recorded. Second, it must be *independently verifiable* — any party can confirm the ordering without relying on the attestation of the party that recorded the events. Third, it must be *globally consistent* — all participants in the substrate agree on the same ordering, even if they are in different jurisdictions, on different networks, or operating at different speeds.

The combination of these properties — unforgeability, independent verifiability, global consistency — is what makes temporal ordering a trust primitive rather than merely a technical feature. It is the digital equivalent of physical causality: once event A is recorded before event B, this ordering becomes a fact of the substrate, not a claim by any participant.

6.5 Permissionless: Any Agent, Any Jurisdiction, No Gatekeepers

A trust substrate for the global agentic internet must be accessible to any agent, in any jurisdiction, without requiring permission from any gatekeeper.

This requirement is a direct consequence of the architectural analysis. If the substrate requires registration, certification, or approval before an agent can use it, then the registrar, certifier, or approver becomes a centralized trust authority — reintroducing the single-point-of-failure and geopolitical problems identified in Section 6.2. If the substrate charges fees that only certain entities can afford, it becomes an exclusionary system that fragments the agent economy along economic lines. If the substrate requires agents to identify their human operators before participating, it reverts to the Era 2 model of human-identity-derived trust — which, as Section 2 demonstrated, is exactly the model that agents break.

Permissionlessness does not mean that agents on the substrate have no identity or accountability. It means that the ability to *establish* identity and *build* reputation does not depend on any gatekeeper's approval. Any agent can participate. But participation leaves verifiable traces — traces that can be used by other agents, by platforms, by enterprises, and by regulators to assess trustworthiness. The substrate does not vouch for any agent. It provides the infrastructure through which agents vouch for themselves — through accumulated, verifiable, tamper-proof records of their behavior over time.

This is a fundamentally different trust model from the one the internet currently uses. In the current model, trust is *delegated* — a trusted third party (a CA, a government, a platform) vouches for an entity, and other parties rely on that vouching. In the substrate model, trust is *earned* — an entity accumulates verifiable evidence of its behavior, and other parties evaluate that evidence directly. The substrate does not decide who is trustworthy. It provides the evidence on which trust decisions can be made.

The permissionless property ensures that this trust model is universal — that any agent, anywhere, can begin building a verifiable track record without first obtaining approval from an authority that may not recognize it, that may not exist in its jurisdiction, or that may deny it access for reasons unrelated to its actual trustworthiness.

6.6 Scarce: Identity and Reputation Must Have Cost

This is perhaps the most counterintuitive property, but it is also the most important for the agent trust problem specifically.

Section 3 demonstrated that the core vulnerability of the current trust architecture is the *cheapness* of digital fabrication. Synthetic identities are dangerous because they cost nothing to create. Fake reviews are dangerous because they cost nothing to write. Fabricated histories are dangerous because they cost nothing to generate. The scarcity that made these trust signals meaningful in Era 1 — the scarcity of human time, human attention, human presence — has been eliminated by AI agents that can generate

unlimited quantities of plausible identity, reputation, and history at the cost of a few cents of compute.

A trust substrate must reintroduce scarcity into the digital realm. Specifically: establishing an identity on the substrate, anchoring an action to the substrate, and building a reputation on the substrate must each require the expenditure of a *real, finite, scarce resource* that cannot be fabricated, counterfeited, or obtained for free.

This is the digital analog of a concept that is intuitive in the physical world. In Era 1, identity was scarce because bodies are scarce. You cannot fabricate a body. Reputation was scarce because time is scarce. You cannot fabricate ten years of lived experience. Accountability was enforced because physical vulnerability is scarce. You cannot escape the jurisdiction of gravity.

In the digital realm, the scarce resource cannot be bodies or time (agents have neither). It must be something else — some resource that is finite, measurable, and physically constrained. The substrate must be designed so that *using* it consumes this scarce resource, creating an inherent cost that prevents the unlimited fabrication of identities, histories, and reputations.

The cost need not be high. It need only be *nonzero and unfakeable*. The difference between zero-cost identity and minimal-cost identity is the difference between an adversary that can create a million fake agents for free and an adversary that must expend real resources for each one. At sufficient scale, even small per-identity costs create meaningful economic barriers to the fabrication attacks described in Section 3.

6.7 Synthesis: A System Whose Security Derives From Physics, Not Promises

The six properties above are not independent requirements. They are facets of a single underlying principle.

The trust substrate for the agentic internet should be a system whose security derives, to the greatest extent achievable, from the laws of physics — from thermodynamic irreversibility, from the conservation of energy, from the physical constraints on computation — rather than from the promises of institutions, the assumptions of mathematics, or the honesty of administrators.

Not all six properties carry equal certainty. Four of them — natively digital (6.1), decentralized (6.2), immutable (6.3), and permissionless (6.5) — follow with high confidence from the problem analysis of Sections 2-5. They are derived directly from specific, documented failure modes. Two of them — temporal ordering (6.4) and physical grounding within Section 6.1 — are strong engineering recommendations rather than logical necessities. A system that satisfied the first four but lacked strict temporal ordering or physical grounding might still provide significant value as agent trust infrastructure, though it would be weaker at the foundation than one that satisfied all six.

Natively digital, physically grounded (6.1): the system operates in cyberspace but its core guarantees come from physics, mediated by cryptographic mechanisms.

Decentralized (6.2): no single institution controls it, because institutional control is a form of promise-based security.

Immutable (6.3): records cannot be altered, not because an administrator promises not to alter them, but because the physical cost of alteration exceeds any plausible economic incentive.

Temporally ordered (6.4): events are sequenced by the substrate itself, not by a trusted clock operator, and this sequencing is enforced by the same physical constraints that enforce immutability.

Permissionless (6.5): any entity can participate, because gatekeeping reintroduces institutional authority

— the very dependency the substrate is designed to eliminate.

Scarce (6.6): identity and reputation have cost, denominated in a physically constrained resource, because cost-free digital objects cannot serve as trust anchors in a world where AI can generate unlimited quantities of anything digital.

The synthesis: what is required is a digital system that imports the trust properties of the physical world — scarcity, irreversibility, temporal ordering — into cyberspace, without requiring the actors in cyberspace to have physical bodies.

This is a precise specification. It narrows the field of candidate systems significantly:

- A centralized database fails on decentralization and immutability (the operator can alter records).
- A permissioned blockchain fails on permissionlessness and decentralization (the consortium controls access).
- A proof-of-stake blockchain satisfies decentralization, immutability, permissionlessness, and temporal ordering, but its security is denominated in its own token rather than in a physical resource — a weaker form of grounding that introduces the circularity concerns discussed in Section 5.4.
- A government-operated ledger fails on decentralization, permissionlessness, and geopolitical neutrality.
- A corporate platform fails on the same dimensions, with the additional risk of commercial incentives distorting trust.

The specification is demanding. The reader is invited to evaluate *any* candidate system — existing or hypothetical — against these six properties. A system that satisfies all six is a strong candidate for the trust substrate. A system that satisfies most but not all may still serve as useful infrastructure in specific domains, but would carry structural limitations as a *universal* trust substrate. Section 7 examines the strongest existing candidate.

References — Section 6

[51] NIST, "Post-Quantum Cryptography Standardization," August 2024. NIST finalized its first set of post-quantum cryptographic standards (FIPS 203, 204, 205), acknowledging that quantum computers could break current public-key cryptography. <https://csrc.nist.gov/projects/post-quantum-cryptography>

7. Bitcoin as the Trust Anchor

Section 6 derived, from first principles, the properties that a trust substrate for the agentic internet must possess: natively digital but physically grounded, decentralized, immutable, temporally ordered, permissionless, and scarce — with the strongest candidate rooting its security in the laws of physics rather than in institutional promises.

This section examines the strongest existing candidate. That system is Bitcoin.

This claim is not made as advocacy. It is made as engineering evaluation — the same way one might assess candidate materials for a bridge foundation by testing each against the load-bearing requirements.

The properties were derived independently of any technology. The technology is evaluated against the properties. The match is either present or it is not.

7.1 Proof-of-Work: Importing Physical Constraints Into Cyberspace

The mechanism by which Bitcoin satisfies the Section 6 requirements is proof-of-work — and understanding proof-of-work correctly is essential to understanding why Bitcoin, specifically, qualifies as a trust substrate.

Proof-of-work is often described as "solving complex mathematical puzzles" or "wasting electricity." Both descriptions miss the point. What proof-of-work actually does is convert a physical resource — electricity — into a cryptographic proof. This conversion is a one-way function: electricity can be converted into proof, but proof cannot be converted back into electricity. The energy is consumed. The proof remains.

This one-way conversion is the bridge between the physical and digital worlds — the mechanism that imports the constraints of physics into cyberspace. To understand why this matters, recall the central paradox identified in Section 5.3: the trust substrate must be natively digital (accessible to entities without physical bodies) and should be as physically grounded as possible (its security should derive from physical constraints rather than institutional promises). Proof-of-work resolves this paradox. The output is purely digital — a block of data, cryptographically linked to the previous block, verifiable by any computer anywhere in the world. But the *cost* of producing that output is physical — measured in kilowatt-hours of electricity, irreversibly consumed. The digital record inherits the physical cost of its creation.

This is not a metaphor. It is a thermodynamic fact. The second law of thermodynamics states that entropy in a closed system never decreases — energy, once consumed, cannot be unconsumed. A Bitcoin block, once mined, represents a quantity of energy that has been permanently converted from usable to unusable form. To alter the block — to change a transaction, to rewrite history — an attacker would need to expend *at least* the same quantity of energy again. The security of the record is denominated not in passwords or promises but in joules.

This is what makes Bitcoin's security qualitatively different from every other form of digital security. Passwords can be guessed. Private keys can be stolen. Database administrators can alter records. Government officials can be bribed or coerced. But no one can un-consume electricity. The laws of thermodynamics are not subject to institutional failure, geopolitical pressure, or regulatory change. They are the most reliable guarantees in the known universe.

An important qualification is necessary here, in the interest of intellectual honesty. The physical cost of Bitcoin's proof-of-work is real and irrecoverable. But the *motivation* for miners to pay that cost is economic: they expect the block rewards and transaction fees they earn to be worth more than the electricity they consume. If that expectation collapses — if BTC loses all value, if mining becomes unprofitable — miners will stop mining, hashrate will fall, and the physical cost barrier will erode. The physics is unconditional; the willingness to pay for the physics is not.

This means Bitcoin's security is not *purely* physics-grounded in an absolute sense. It is physics-grounded *conditional on* continued economic participation. This is a weaker claim than "thermodynamic guarantee" — but it is a substantially stronger claim than any alternative offers. The distinction: in a promise-based system, security depends on institutional integrity (a single point of failure). In a proof-of-stake system, security depends on token value (a self-referential loop). In Bitcoin, security depends on the ongoing economic incentive to expend a physical resource — a dependency that is simpler, more robust, and has held without interruption for sixteen years under extreme adversarial pressure. Bitcoin provides the strongest available *approximation* of physics-grounded security — the closest any digital system has come to importing unconditional physical constraints into cyberspace.

The question for the trust substrate is not whether Bitcoin's physical grounding is *absolute* (it is not), but whether it is the *most physically grounded* option available (it is). For a root of trust, the engineering choice is the option with the strongest foundation, not the option with a theoretically perfect one.

To restate in the language of Section 6: proof-of-work is the mechanism by which Bitcoin achieves the Property 6.1 requirement — a system that is natively digital in its interface but physically grounded in its security, mediated by an economic incentive structure that has proven remarkably durable. Every other property follows from this foundation.

7.2 The Three Properties That Proof-of-Work Imports

Section 5.3 identified three properties that physics provided naturally in Era 1 — scarcity, irreversibility, and temporal ordering — and that the digital world lacks. Proof-of-work reimports all three.

Scarcity. In the physical world, an object cannot be in two places simultaneously. This constraint prevents double-spending of physical assets: if you hand someone a gold coin, you no longer have it. In the digital world, data can be copied infinitely — a purely digital asset can be "spent" simultaneously to a thousand counterparties. Bitcoin's proof-of-work consensus mechanism solves this by requiring the network to agree, through competitive energy expenditure, on a single canonical history of transactions. A bitcoin (the unit of value) cannot be double-spent because the network's miners — collectively expending hundreds of terawatt-hours of energy — enforce a single, authoritative ledger. The scarcity is not asserted by an institution. It is enforced by the energy expenditure of the network.

This scarcity property directly addresses the requirement identified in Section 6.6: that identity and reputation on the trust substrate must have *cost*. On Bitcoin, every operation — anchoring data, establishing identity, building a verifiable history — requires the expenditure of a real, finite resource. This expenditure cannot be fabricated, counterfeited, or bypassed. An adversary seeking to create a million fake identities must pay the cost of anchoring a million identities — a cost denominated in the same scarce resource (energy, translated to bitcoin transaction fees) that secures the entire network.

Irreversibility. In the physical world, actions cannot be undone. A building once demolished cannot be undemolished. A bullet once fired cannot be unfired. This irreversibility is what makes physical accountability possible: because physical actions have permanent consequences, actors must bear responsibility for them. In the digital world, records can be altered, transactions can be reversed, and history can be rewritten — which is precisely why digital accountability has always depended on institutional enforcement rather than architectural guarantee.

Bitcoin's proof-of-work creates digital irreversibility. Once a transaction is confirmed and buried under subsequent blocks — each representing additional energy expenditure — reversing it requires re-expending all of the energy consumed by every subsequent block. As of late 2025, the Bitcoin network consumes approximately 150 terawatt-hours per year.⁵² The energy embedded in the chain grows with every ten-minute block. Reversing even a single day's worth of transactions would require energy expenditure on a scale that exceeds the output of many nations' entire electrical grids. This is constraint-based immutability — the Property 6.3 requirement — in its most literal form.

Temporal ordering. In the physical world, time flows in one direction. Event A happens before event B, and this ordering cannot be retroactively changed. This temporal ordering is what makes physical trust relationships possible: contracts are enforceable because the sequence of offer, acceptance, and performance is fixed in time. In the digital world, timestamps can be falsified and event sequences can be rearranged — which is why digital trust has always required trusted third parties (timestamp authorities, database administrators, platform operators) to attest to the ordering of events.

Bitcoin's blockchain is, at its core, a temporally ordered ledger. Each block contains a cryptographic hash

of the previous block, creating a strict, one-directional chain. The ordering of blocks — and therefore the ordering of all events recorded within them — is enforced by the proof-of-work consensus: the valid chain is the one with the most cumulative work, and cumulative work can only increase in one direction. No party can insert a block retroactively, remove a block from the middle of the chain, or reorder the sequence of events without re-expending the energy of every subsequent block. The temporal ordering is not asserted by a timestamp authority. It is enforced by thermodynamics.

This directly satisfies the Property 6.4 requirement: a strict, unforgeable, independently verifiable, globally consistent sequence of events — the digital equivalent of physical causality.

7.3 Why Proof-of-Work and Not Proof-of-Stake

A reasonable objection at this point is: "Other blockchains exist. Why specifically Bitcoin? Why not Ethereum, or Solana, or any other decentralized ledger?"

The answer lies in the distinction between proof-of-work and proof-of-stake, and it is an engineering distinction — one of degree rather than category, but a degree that matters enormously for a root of trust.

Proof-of-stake (PoS) secures a blockchain by requiring validators to deposit ("stake") the blockchain's native token as collateral. Validators who behave dishonestly have their stake "slashed" — confiscated by the protocol. The security of the chain depends on the economic cost of acquiring enough stake to control the network.

Proof-of-work (PoW) secures a blockchain by requiring miners to expend electricity — a physical resource — to produce valid blocks. The security of the chain depends on the energy expenditure required to overwrite its history.

Both systems involve economic incentive loops. In PoW, miners mine because they expect BTC rewards and fees to be worth more than their electricity costs; if that expectation collapses, security erodes. In PoS, validators stake because they expect the token to maintain value; if that expectation collapses, security erodes. *Neither system provides unconditional security.* Both depend on continued economic participation.

The difference — and it is a difference that matters for a root of trust — lies in the *denomination* of the cost of attack and the *structure* of the dependency.

In a PoS system, the cost of attack is denominated in the chain's own token. The token's value depends on the market's confidence in the chain's security. The chain's security depends on the total value staked. This creates a *self-referential* dependency: security → token value → stake value → security. If confidence drops, token value falls, reducing the cost of attack, which further reduces confidence. The system has two layers of economic contingency: miners/validators must expect profit, *and* the token must maintain value for the staking incentive to function. The reflexive doom loop is a structural vulnerability — not a theoretical one, but one that has been observed in practice when PoS chains experience rapid devaluation.

In a PoW system, the cost of attack is denominated in an *external* physical resource: energy. Electricity prices are set by global energy markets, semiconductor supply chains, and the laws of physics — not by the token's market price. An attacker must acquire physical hardware that takes months to manufacture and physical electricity that is consumed irrecoverably. The system has one layer of economic contingency: miners must expect profit from mining. But the cost already paid — the energy already consumed — is sunk. It cannot be reflexively devalued. The history already secured remains secured by the energy already spent, regardless of what happens to the token's price tomorrow.

This is a degree difference, not a category difference. Both PoW and PoS involve economic participation. But the degree difference is substantial and structural, not marginal. PoW's economic contingency is

simpler (one layer vs. two), externally anchored (energy markets vs. self-referential token value), cumulative and irreversible (energy already spent secures past records permanently, regardless of future token price) vs. reflexive and reversible (token devaluation degrades security of the entire chain simultaneously, past and present). And PoW is empirically more robust — sixteen years of unbroken operation vs. no PoS system operating at comparable scale for comparable duration under comparable adversarial pressure. These are not marginal differences. They are architectural differences in the dependency structure of the security model.

For *application-layer computation*, this difference may not matter. Ethereum and other PoS chains provide valuable computation platforms and have legitimate use cases. But for a *root of trust* — the foundational, bedrock layer from which all other trust derives — the engineering case favors the system with the simplest dependency structure and the strongest empirical record. A root of trust, by definition, should depend on as few assumptions as possible. PoW depends on one economic assumption (mining profitability) and one physical constraint (energy expenditure). PoS depends on two economic assumptions (validator profitability and token value) with no physical constraint. For the foundation of the global agent economy, fewer and stronger dependencies are preferable.

7.4 Bitcoin's Track Record: Sixteen Years of Unbroken Security

Theory is necessary but insufficient. A trust substrate that is theoretically sound but practically untested is not a trust substrate — it is an experiment. The root of trust for the global agent economy must be *proven*, not merely designed.

Bitcoin has been operational since January 3, 2009. In the sixteen years since its creation, the base-layer protocol has never been successfully attacked. No 51% attack has ever been executed against the Bitcoin network.⁵³ No transaction confirmed on the Bitcoin blockchain has ever been reversed by an adversary. No coins have been counterfeited. No blocks have been fraudulently inserted.

This is not because no one has tried. Bitcoin is, by a considerable margin, the most economically incentivized system in history to attack. As of early 2026, the network secures hundreds of billions of dollars in value. It has been continuously probed by state-level adversaries, criminal organizations, academic researchers, and opportunistic hackers for over a decade and a half. Every conceivable attack vector — 51% attacks, selfish mining, eclipse attacks, time-warp attacks — has been analyzed, simulated, and, in the case of smaller proof-of-work chains, executed against weaker networks. Bitcoin's base layer has withstood all of them.

The cost of attack provides a quantitative measure of this resilience. Coin Metrics' 2024 research, the first rigorous empirical analysis of blockchain attack costs, found that a 51% attack on Bitcoin would require acquiring approximately 7 million ASIC mining rigs — a quantity that simply does not exist on the global market — at a cost exceeding \$20 billion for hardware alone.⁵⁴ A more recent analysis by Duke University's Professor Campbell Harvey estimated the total cost of a one-week sustained 51% attack at approximately \$6 billion, comprising \$4.6 billion in hardware, \$1.34 billion in data center construction, and \$130 million in weekly electricity costs.⁵⁵ Both analyses concluded that no profitable attack strategy exists — even in the most favorable scenario, the attacker's potential return would be a fraction of the expenditure required.⁵⁴

Critically, this security is *cumulative*. Every block mined adds to the total energy embedded in the chain. The cost of attacking any specific historical record grows with every subsequent block, because an attacker must re-expend not just the energy of the target block but the energy of every block that follows it. Bitcoin's security does not remain constant over time — it *increases*, monotonically, with every ten-minute block. The chain is not just secure today. It is more secure today than it was yesterday, and it will be more secure tomorrow than it is today.

This track record matters for the trust substrate argument because trust substrates are, by nature, conservative infrastructure. The trust layer of the global economy cannot be an untested innovation. It must be a battle-tested system with a demonstrated record of resilience against the most sophisticated attacks. Bitcoin's sixteen-year record of unbroken security, maintained against continuous adversarial pressure, is not just impressive — it is the empirical foundation without which the theoretical argument of this paper would be insufficient.

7.5 The Fat Protocol: Why the Network and the Token Are Inseparable

There is a common misconception that Bitcoin's trust properties could be abstracted away from Bitcoin the asset — that one could use "blockchain technology" for trust anchoring without engaging with BTC the token. This misconception is important to address because it has practical implications for how the trust substrate functions and how value flows through it.

In the early internet, value accrued primarily to the *application* layer, not the *protocol* layer. TCP/IP, HTTP, and SMTP are open protocols that anyone can use for free. The value they created was captured by the companies built on top of them — Google, Amazon, Facebook. This is why TCP/IP, despite being the most widely used protocol in human history, has no associated market value. The protocol is free. The applications are valuable.

Bitcoin inverts this structure. The protocol (the Bitcoin network) and the scarce asset (BTC) are fused together at the architectural level. You cannot use Bitcoin's trust properties — its immutability, its temporal ordering, its constraint-based security — without interacting with BTC. Anchoring data to the Bitcoin blockchain requires paying a transaction fee in BTC. The miners who secure the network are compensated in BTC. The economic incentive that drives miners to expend energy — the energy that provides the physical grounding of the entire trust system — is denominated in BTC.

This fusion has a critical consequence: *adoption of Bitcoin's trust properties drives demand for BTC, which increases miner revenue, which attracts more mining power, which increases the network's security, which makes the trust properties stronger*. This is a reflexive loop — a positive feedback cycle in which the trust substrate becomes *more secure* as more value flows through it.

A careful reader will note that this reflexive loop has the same *logical form* as the economic dependency identified in Section 7.3's analysis of proof-of-stake: adoption → value → security → adoption. This is an important observation, and intellectual honesty requires addressing it directly rather than eliding the structural similarity.

The difference lies not in the form of the loop but in its *content* — specifically, in what happens at the "value → security" step. In Bitcoin's loop, increased value translates to increased miner revenue, which translates to increased *physical energy expenditure*. The security gained at each turn of the loop is physically embedded in the chain — the energy has been consumed, the work has been done, and it cannot be reflexively unwound. If Bitcoin's value drops tomorrow, the security of *past* blocks is unaffected because the energy securing them has already been spent. The loop strengthens security cumulatively and irreversibly.

In a PoS loop, increased value translates to increased staking, which translates to a higher *nominal cost of attack denominated in the same token*. If the token's value drops, the security of the entire chain — past and present — degrades simultaneously, because the cost of acquiring enough stake to attack falls in real terms. The loop can unwind as quickly as it wound up.

This is not a minor distinction. It is the difference between a system whose security *accumulates* over time (each turn of the loop adds permanently) and a system whose security can *evaporate* (each turn of the loop is only as durable as the current token price). For a trust substrate that must maintain the integrity of

records anchored years ago, cumulative irreversible security is an essential property.

Compare this to TCP/IP. If a billion new users start using TCP/IP tomorrow, the protocol does not become more reliable. Its capacity may strain, but its *security* properties are unchanged. TCP/IP does not become harder to attack because more people use it.

If a billion new agents start anchoring their identities and actions to the Bitcoin blockchain, the network *does* become more secure. The increased demand for block space increases transaction fees, which increases miner revenue, which attracts more hashrate, which increases the cost of attack. The trust substrate strengthens in direct proportion to its adoption. This is not a bug — it is a design property that aligns the interests of every participant in the ecosystem. Every agent that uses Bitcoin for trust anchoring contributes to the security that protects every other agent.

For the trust substrate argument, this means that Bitcoin's suitability is not static. It improves with adoption. The more agents use it, the stronger it gets. This is the opposite of most infrastructure, which degrades under load. Bitcoin's trust properties improve under load — because load translates to revenue, revenue translates to mining investment, and mining investment translates to energy-backed security.

7.6 What Bitcoin Is Not: Clarifying the Scope

It is essential, having made the case for Bitcoin as the trust anchor, to be equally precise about what Bitcoin is *not*.

Bitcoin is not the *execution layer* for agent interactions. Agents will not conduct their moment-to-moment business on the Bitcoin base chain. Bitcoin processes approximately seven transactions per second on its base layer. The global agent economy will generate millions of interactions per second. Bitcoin cannot and should not handle this volume directly.

Bitcoin is not the *application layer*. It will not provide agent identity interfaces, reputation dashboards, or compliance tools. These are application-level concerns that will be built by companies, open-source projects, and ecosystems operating above the trust substrate.

Bitcoin is not the *entire solution*. It provides the irreducible foundation — the root of trust — upon which a complete trust architecture is built. The analogy is geological. Bitcoin is the bedrock. The buildings — the identity systems, the audit platforms, the reputation networks, the compliance frameworks — are built on top of the bedrock. The bedrock does not determine the architecture of the buildings. But the structural integrity of every building depends, ultimately, on the solidity of the bedrock beneath it.

The layered architecture works as follows. The Bitcoin base layer provides timestamping, ordering, and immutability — the three base-layer operations from which all higher-layer trust mechanisms are constructed, and which no other layer can provide as securely. Above this, protocol layers like the Lightning Network provide speed and micropayment capability. Above this, application protocols provide batching (aggregating thousands of agent actions into a single Bitcoin transaction via Merkle trees), identity management, and reputation computation. Above this, user-facing applications provide the interfaces through which enterprises, developers, and agents interact with the trust substrate.

Each layer inherits the trust properties of the layer below it. An identity record anchored to Bitcoin via a Merkle root inherits Bitcoin's immutability. A reputation score derived from on-chain history inherits Bitcoin's temporal ordering. An audit trail batched and anchored to the blockchain inherits Bitcoin's constraint-based security. The trust flows upward, from the bedrock to the application, and it flows because — and only because — the bedrock is sound.

This layered model means that the Bitcoin base layer's throughput limitation is not a constraint on the agent trust architecture. It is a design feature. The base layer optimizes for security and decentralization,

not speed. Speed is handled by higher layers. The result is a system that provides the strongest possible trust guarantees at the foundation while enabling the speed and scale that agents require at the application layer.

Section 8 will describe this layered architecture in concrete terms — showing exactly how agent identity, action auditing, and reputation can be built on the Bitcoin trust anchor using technology that already exists.

References — Section 7

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8. How Bitcoin Anchors Agent Trust — A Concrete Architecture

The previous sections established *why* a trust substrate is needed and *why* Bitcoin satisfies the requirements. This section addresses the practical question: *how*? How do you actually build agent identity, action auditing, and reputation on top of a blockchain that processes seven transactions per second?

The answer is a layered architecture — one that uses Bitcoin as the root of trust for anchoring, not as the execution environment for every agent interaction. The building blocks for this architecture already exist. Several have been operational for years. What is missing is not technology but *integration* — the assembly of existing components into a coherent trust infrastructure for autonomous agents.

This section is deliberately architectural rather than implementational. The goal is to demonstrate feasibility and convey the core design, not to provide a complete technical specification. A full implementation reference is a separate document.

8.1 Agent Identity: Cryptographic Birth Certificates Anchored to Bitcoin

The first primitive is identity. In the current internet, agent identity is a blank — an API key, an OAuth token, a session cookie. None of these carry information about who created the agent, when it was created, what model it runs, what permissions it was designed to have, or what constraints govern its behavior. When an agent misbehaves, investigators must trace backward through logs, infrastructure, and organizational records to determine its provenance — a process that takes days or weeks, if it succeeds at all.

A Bitcoin-anchored identity system works differently.

When an agent's creator — a company, a developer, an AI platform — instantiates a new agent, they generate a cryptographic certificate: a structured data object containing the agent's public key, the creator's identity (itself anchored to a prior Bitcoin commitment), the agent's declared capabilities and constraints, the model version and configuration hash, and the timestamp of creation. This certificate is the agent's *identity document* — its digital birth certificate.

The creator then computes the cryptographic hash of this certificate — a unique digital fingerprint — and anchors that hash to the Bitcoin blockchain. Anchoring means embedding the hash in a Bitcoin transaction such that it becomes part of the immutable, temporally ordered, globally verifiable record. Once confirmed, this anchor proves three things:

Existence: This specific agent, with these specific properties, existed at or before this specific time. The proof is mathematical, not testimonial — anyone with the certificate and the blockchain can verify it independently.

Provenance: This agent was created by this specific entity. The creator's signature on the certificate, combined with the Bitcoin anchor, creates an unbreakable link between agent and creator. The creator cannot later deny having created the agent (non-repudiation), and no one else can fraudulently claim to have created it.

Integrity: The agent's properties have not been altered since creation. If anyone modifies the certificate — changes the declared capabilities, alters the model version, adjusts the constraints — the hash changes, and the modification is detectable. The original, anchored version is the canonical one.

If the agent's creator needs to *update* the certificate — because the model was upgraded, the permissions changed, or the agent was decommissioned — they issue a new certificate referencing the previous one, and anchor that new certificate to Bitcoin. This creates a verifiable *chain of identity* — an auditable history of every change to the agent's identity and properties, ordered in time, anchored to the immutable ledger. Revocation works the same way: the creator publishes a signed revocation notice, anchors it, and any party evaluating the agent's identity can verify that it was revoked at a specific time.

The result is an identity system where every agent has a verifiable, tamper-proof provenance record. No central registry is required. No government issues the certificate. No platform mediates the verification. Any party — another agent, an enterprise, a regulator, a court — can independently verify the identity by checking the certificate against the blockchain. The trust is mathematical, not institutional.

An important operational caveat: Bitcoin-anchored identity provides strong guarantees about *historical* identity — who created this agent, when, with what properties, and whether the identity has been revoked. It does not, by itself, solve the *real-time* identity problem: verifying that the signing key is still controlled by its legitimate holder at the moment of a specific interaction. If an agent's private key is compromised — through cloud breach, supply-chain attack, or malware — an attacker can produce validly signed actions under the victim's identity until revocation propagates. This key-compromise-and-revocation-latency problem is inherent to every identity system ever built, physical or digital: passports can be stolen, PKI certificates can be compromised between CA revocation checks, and bearer credentials can be misused before cancellation. The trust substrate does not eliminate this problem. It mitigates it

through tamper-proof revocation records (enabling precise forensic determination of which actions occurred before vs. after compromise), and it provides a foundation on which operational mitigations — short-lived certificates, OCSP-style stapled freshness proofs, threshold signature schemes that require multiple key shares to sign, and Lightning-speed revocation broadcast — can be built at higher layers. The substrate provides the evidentiary bedrock; real-time key freshness is a Layer 3 operational responsibility.

A related but distinct failure mode: what happens if the creator *loses* the private key entirely? An important distinction applies here. The Bitcoin-anchored identity certificate is a *record* of the agent's identity and provenance — it is not the *operational control mechanism* for the agent. The agent runs on infrastructure (cloud servers, enterprise clusters, compute platforms), and the creator retains operational access through standard infrastructure controls — they can shut down the container, revoke OAuth tokens, terminate the process. What the lost key prevents is the ability to *cryptographically communicate that change to the trust substrate* — to sign a revocation notice that other parties can independently verify against the original anchored certificate. The agent is operationally dead, but on the trust substrate its identity certificate appears valid and unrevoked.

The dangerous variant of this scenario is when the creator loses *both* the private key and operational access to the agent's infrastructure — because the agent has migrated to decentralized compute, a cloud provider is uncooperative, or the agent was designed with sufficient autonomy to provision its own resources. Now a potentially rogue agent operates with a valid, unrevocable identity certificate, and no cryptographic mechanism exists to signal that it should no longer be trusted.

This failure mode is real but solvable through well-established cryptographic and operational practices, several of which should be incorporated into any production deployment of this architecture. *Threshold signatures (M-of-N keys)* distribute signing authority across multiple key shares held by different people, devices, or locations — losing one or even two shares does not lock the creator out of revocation authority, a pattern already standard in Bitcoin multisig custody. *Pre-signed emergency revocations* are revocation transactions signed at the time of agent creation but not broadcast — held in reserve by the creator, legal counsel, or a compliance service, deployable even if the creator's active key is subsequently lost, analogous to Bitcoin's timelocked transactions used in Lightning channel dispute resolution. *Certificate expiry with mandatory renewal* builds a finite validity period (e.g., 90 days) into each certificate, requiring the creator to periodically re-sign and re-anchor; if the key is lost, the certificate expires by default rather than persisting indefinitely, converting key loss from a permanent problem to a bounded one. *Social recovery mechanisms* at higher layers allow the creator's own identity — a corporate entity, a multisig governance structure, or a legal representative — to issue superseding revocations through alternative authentication channels when the primary key is unavailable.

No single mitigation eliminates the risk entirely. In combination, they reduce the window of vulnerability from permanent to bounded, and they ensure that key loss — while operationally disruptive — does not create permanently unrevocable rogue agent identities on the trust substrate.

8.2 Action Audit Anchoring: Proving What Happened and When

Identity establishes *who* an agent is. The second primitive establishes *what it did*.

When an agent performs actions — executing transactions, accessing databases, generating content, communicating with other agents, making decisions — those actions can be logged. This is not new; enterprises already log agent activity. What is new is the ability to make those logs *independently verifiable* — to prove, to any third party, that the log has not been altered since it was created.

The mechanism is Merkle tree batching with Bitcoin anchoring, and it has been operational since 2016 through the OpenTimestamps protocol.⁵⁶

Here is how it works:

Step 1: Log. The agent (or the infrastructure it runs on) records actions in a structured log. Each log entry contains the action taken, the timestamp, the agent's identity (linked to its Bitcoin-anchored certificate), and any relevant context — counterparty identity, data accessed, decision rationale. The specific logging schema — what constitutes an "action," what context must be recorded, and how entries are structured for interoperability — is an application-layer standardization challenge, not a substrate concern. This is analogous to how TCP/IP provides reliable data transport without specifying what applications should transmit; the trust substrate provides tamper-proof anchoring without specifying what should be anchored. Standardization bodies, regulatory requirements (such as the EU AI Act's logging mandates), and industry consortia will define logging schemas appropriate to their domains. The substrate is agnostic to schema content — it anchors whatever is submitted and guarantees its integrity.

Step 2: Hash. Each log entry is cryptographically hashed — converted into a fixed-length digital fingerprint that uniquely identifies the entry's contents. If even a single bit of the entry is changed, the hash changes completely.

Step 3: Batch into a Merkle tree. Hundreds, thousands, or millions of hashed log entries are organized into a Merkle tree — a hierarchical data structure in which each parent node is the hash of its children. The tree has a single root hash at the top — a single value that uniquely represents the *entire* batch of entries. If any entry anywhere in the tree is altered, the root hash changes.

Step 4: Anchor the root. The Merkle root — a single 32-byte value — is embedded in a Bitcoin transaction. One Bitcoin transaction, costing a few dollars in fees, anchors an arbitrary number of log entries.

Step 5: Verify. At any future time, any party can verify a specific log entry using only three things: the entry itself, a compact *Merkle proof* (a small set of sibling hashes along the path from the entry to the root — typically around 20 hashes for a tree of a million entries, totaling less than a kilobyte regardless of tree size), and the Merkle root anchored in the Bitcoin blockchain. The verifier hashes the entry, combines the result with each sibling hash in sequence climbing toward the root, and checks whether the final computed value matches the anchored root. If it matches, the entry is provably unaltered and existed at or before the time the Bitcoin block was confirmed. If it does not match, tampering has occurred. Crucially, the verifier does not need access to any other log entry in the tree — only the entry being verified and its proof. Verification is logarithmic, not linear: checking one entry out of a million requires approximately twenty hash computations, not a million.

The verification requires no trust in any third party. The verifier does not need to trust the entity that created the log, the entity that operates the server, or any intermediary. They need only the log entry, the Merkle proof, and access to the Bitcoin blockchain — which is public, replicated across thousands of nodes worldwide, and independently verifiable by anyone running a Bitcoin node.

OpenTimestamps demonstrates the real-world viability of this approach. The protocol has been operational for nearly a decade, uses public calendar servers that batch thousands of timestamps into a single Bitcoin transaction, and is entirely open source.⁵⁶ The OriginStamp system, a related implementation, has processed nearly two million timestamps using just over a thousand Bitcoin transactions — a batching ratio of approximately 2,000 to 1.⁵⁷ At scale, the marginal cost per timestamp approaches zero.

For the agent trust use case, this means: every high-stakes agent action — every financial transaction, every contract execution, every data access, every clinical recommendation, every procurement decision — can be anchored to Bitcoin at negligible cost. The result is an audit trail whose entries are *individually proven* — proven to be unaltered and proven to have existed in a specific order at a specific time. This is

the "immutable audit trail" that the EU AI Act requires for high-risk AI systems⁵⁸ — implemented with constraint-based immutability (guaranteed by physics) rather than promise-based immutability (guaranteed by a database operator's policies).

Two important limitations must be stated explicitly. **Limitation 1: Truthfulness (the oracle problem).** Bitcoin-anchored audit trails prove that a record existed at a specific time and has not been altered since. They do *not* prove that the record is *truthful*. If an agent logs a lie — recording that it performed action A when it actually performed action B — that lie will be anchored immutably. The Merkle proof will confirm the lie has not been tampered with. The oracle problem — the problem of verifying that data fed into a system accurately reflects external reality — is not solved by the trust substrate.

Limitation 2: Completeness. Anchoring proves the integrity of *what was recorded*, not that *everything that happened* was recorded. A malicious agent — or an adversary who has compromised an agent via prompt injection, model poisoning, or supply-chain attack — can selectively omit incriminating actions from the log. The resulting Merkle tree is mathematically pristine and provably unaltered. It is also incomplete. This is a fundamental constraint of any commitment scheme: it binds the commitment to the data submitted, not to the universe of data that should have been submitted.

These limitations are real, but they do not undermine the value of tamper-evidence for accountability. Consider what happens when the truth later emerges. In a system with promise-based logs, the agent (or its operator) can retroactively alter the log to match the newly revealed truth, destroying evidence of the original lie. In a system with Bitcoin-anchored logs, the original lie is permanently recorded. The agent cannot revise its story. The discrepancy between the anchored log and the truth is itself evidence — evidence that the agent or its operator recorded false information, which is an actionable finding for regulators, courts, and counterparties.

Tamper-evidence does not guarantee truth or completeness. It guarantees that *the story cannot change after the fact*. For accountability, this is a critical property. People and organizations rarely face consequences for the original misdeed alone — they face consequences for the cover-up, for the altered records, for the discrepancy between what they said happened and what actually happened. An immutable audit trail makes cover-ups impossible. That is its contribution to the accountability stack described in Section 5.5.

The completeness gap, while not eliminable at the substrate level, can be substantially narrowed through architectural measures described in Section 8.6.

8.3 Agent Reputation: Verifiable Track Records Without Central Authority

Identity establishes who an agent is. Audit anchoring establishes what it did. The third primitive is *reputation* — a mechanism by which agents that have never interacted can assess each other's trustworthiness.

In the current internet, reputation is centrally mediated. Amazon provides seller ratings. Uber provides driver ratings. Credit bureaus provide credit scores. In each case, a central authority aggregates behavioral data and publishes a score. The score is trusted because the authority is trusted. The authority is trusted because it is a large, regulated institution with a reputation of its own. This is Era 2 trust — institutional, centralized, and dependent on the integrity of the mediating organization.

For autonomous agents operating at scale across jurisdictions and platforms, centralized reputation is insufficient for the same reasons identified in Section 4: no single entity is globally trusted, centralized systems are single points of failure, and centralized scores can be manipulated by the entity that controls them.

Bitcoin-anchored identity and audit trails enable a fundamentally different approach: *derived reputation*. Rather than a central authority publishing a score, reputation is computed from the publicly verifiable record of an agent's behavior.

An agent's Bitcoin-anchored history provides the raw data: the age of its identity certificate (how long has this agent existed?), the consistency of its audit trail (has it been continuously anchored, or are there gaps that suggest selective logging?), the volume and pattern of its transactions (is this a new agent with no history, or a seasoned one with thousands of verified interactions?), whether its identity has ever been revoked or modified (has the creator changed its permissions, which might indicate a security concern?), and the reputation of the entities it has interacted with (does it transact primarily with other well-established agents, or with newly created identities that might be fabricated?).

None of this data requires a central aggregator. It is all derivable from the public Bitcoin blockchain and the associated Merkle proofs. Any party can compute it independently. Different parties can weight the factors differently based on their own risk tolerance. An enterprise evaluating a potential counterparty agent might weigh identity age heavily. A high-frequency trading system might weigh transaction volume. A regulatory auditor might weigh audit trail consistency. The raw data is the same — the interpretation is application-specific.

This is the earned trust model described in Section 6.5: the substrate does not decide who is trustworthy. It provides the verifiable evidence on which trust decisions are made. The reputation is not *assigned* by an authority. It is *accumulated* through behavior, recorded immutably, and evaluated independently.

A Sybil defense is essential. A well-funded adversary could create millions of agent identities at low cost (as Section 8.5 demonstrates, batched identity anchoring approaches fractions of a cent per identity), simulate interactions among them, and fabricate reputation networks that appear legitimate. This is a serious attack vector that the reputation system must address.

The defense operates at two levels. First, *identity creation cost* provides a baseline deterrent. Even at fractions of a cent per identity, creating a million fake agents costs thousands of dollars — not prohibitive for a state-sponsored attacker, but sufficient to deter casual fabrication. Second, and more importantly, *reputation-relevant actions must carry meaningful economic commitment*. The tiered cost model works as follows: establishing a bare identity on the substrate is cheap (batched anchoring). But performing actions that contribute to reputation — executing transactions, posting bonds, anchoring high-stakes audit trails — requires economic commitments that scale with the significance of the action. A reputation algorithm that weights the *economic value committed* in anchored actions, not merely the *existence* of actions, is resistant to Sybil fabrication. An adversary can create a million identities cheaply, but building reputation for each one requires committing real economic resources through each one — resources that scale linearly with the number of fabricated reputations.

This tiered model — cheap to *exist*, expensive to *earn trust* — mirrors the physical world. Anyone can walk into a city (permissionless entry). But building a business reputation requires years of verifiable economic activity. The substrate enables the same dynamic digitally: permissionless participation, but reputation earned only through verifiable, economically committed behavior.

The network effects of this system are powerful. Every agent that establishes a Bitcoin-anchored identity and maintains an anchored audit trail adds data to the global reputation graph. The more agents participate, the richer the data, the more meaningful the reputation signals, and the more valuable participation becomes. This creates a positive feedback loop: early adopters benefit from establishing reputation before competition, which attracts more participants, which makes the reputation system more comprehensive, which attracts still more participants.

8.4 The Layered Model: Bitcoin as Root, Everything Else Above

The three primitives described above — identity, auditing, reputation — do not operate directly on the Bitcoin base layer. They are built in a layered architecture that mirrors the internet's own layered design.

Layer 0: Bitcoin base chain. This layer provides exactly three things: timestamping (proving that data existed at a specific time), ordering (proving that event A occurred before event B), and immutability (proving that data has not been altered). It does not process agent interactions. It does not store agent data. It anchors *proofs* — compact cryptographic commitments (Merkle roots) that represent the integrity of data stored elsewhere. This layer is slow (one block every ten minutes), expensive (transaction fees), and maximally secure.

Layer 1: Payment and verification channels. Protocols like the Lightning Network operate above the base chain, providing fast, cheap transactions that settle to the base layer. For agent trust, this layer serves two functions. First, it enables micropayment authentication — agents can prove their willingness to commit economic resources (a scarce signal) through Lightning-speed payments, without waiting for base-layer confirmation. Second, Lightning's L402 protocol (formerly LSAT) provides a native mechanism for machine-to-machine payment authentication — a way for agents to prove they have paid for access to a resource, without human intervention, in milliseconds.⁵⁹

Layer 2: Batching and aggregation. This layer collects agent actions, organizes them into Merkle trees, and periodically anchors the roots to the Bitcoin base chain. OpenTimestamps and similar protocols operate at this layer. The key design principle is *asymmetric scaling*: the number of agent actions that can be verified grows without bound, while the demand on Bitcoin block space remains constant (one Merkle root per batch, regardless of batch size). This is what makes the architecture economically viable at the scale of millions of agents generating billions of actions.

Layer 3: Application protocols and SDKs. This layer provides the developer-facing interfaces — the APIs, libraries, and tools through which agent platforms, enterprise systems, and compliance tools interact with the trust infrastructure. An agent platform like LangChain or CrewAI would integrate a trust SDK that handles identity certificate generation, action logging, Merkle tree construction, Bitcoin anchoring, and reputation querying. The developer does not need to understand Bitcoin internals. They call an API: `anchor_action(action_data)`, `verify_audit_trail(agent_id, time_range)`, `get_reputation_score(agent_id)`.

Layer 4: User-facing applications. Dashboards for enterprise compliance officers to monitor agent audit trails. Interfaces for regulators to verify that AI systems meet EU AI Act requirements. Tools for developers to check the reputation of third-party agents before integrating them. Consumer-facing trust indicators — the equivalent of the browser padlock icon, but for agent interactions.

Each layer inherits the trust properties of the layer below it. The application layer inherits the security of the batching layer. The batching layer inherits the immutability of the base chain. The base chain's security derives from proof-of-work — from thermodynamics. Trust flows upward from physics to user interface, and it flows because every layer is cryptographically linked to the layer below it.

This architecture mirrors the internet's own design: physical layer (cables, radio) → transport layer (TCP/IP) → application layer (HTTP) → user interface (browsers, apps). The internet's power comes from this separation of concerns — each layer does one thing well and provides a clean interface to the layer above. The agent trust architecture follows the same principle.

A design note on this separation of concerns: the higher layers of this architecture (Sections 8.6's corroboration, gap detection, witness redundancy, and economic bonding) are deliberately designed with *ledger-agnostic interfaces*. Their APIs and protocols do not hardcode Bitcoin-specific operations. They

require only the abstract primitives that Layer 0 provides: immutable commitment, consistent global ordering, and public verifiability. This is good engineering — it allows the architecture to be portable in principle while being optimized in practice.

But interface-agnosticism does not mean security-agnosticism. The higher layers inherit the *security ceiling* of the root. If the commitment service at the bottom can be rewritten (PoS reflexive devaluation), corrupted (institutional operator collusion), or fragmented (multiple inconsistent ordering authorities), then every corroboration proof, every cadence check, every witness verification built on top inherits that vulnerability. The choice of root is not a pluggable implementation detail — it is the decision that determines the maximum security any higher-layer mechanism can achieve.

Bitcoin provides the highest available security ceiling: cumulative irreversible energy-backed immutability, sixteen years of unbroken operation, a single consistent global ordering enforced by thermodynamics. The architecture works on other commitment services in the same way that a steel-frame building "works" on a sand foundation — structurally, the interfaces connect, but the load-bearing capacity is determined by what is at the bottom. For the trust infrastructure of the global agent economy, the engineering recommendation is to build on the strongest available foundation.

8.5 Cost and Scalability: Why This Is Economically Viable

The most common objection to Bitcoin-anchored trust infrastructure is cost: "Bitcoin transactions are expensive. You can't put every agent action on the blockchain."

This objection is based on a misunderstanding of the architecture. The design does not put every action on the blockchain. It puts one Merkle root on the blockchain per batch — and a batch can contain an arbitrary number of actions.

Consider the economics concretely.

A single Bitcoin transaction, costing approximately \$1-5 in fees at current rates, can anchor one Merkle root. That Merkle root can represent:

- 1,000 agent actions → cost per action: \$0.001 to \$0.005
- 10,000 agent actions → cost per action: \$0.0001 to \$0.0005
- 1,000,000 agent actions → cost per action: \$0.000001 to \$0.000005

At the scale of millions of agents generating billions of daily actions — the scale the agentic economy is projected to reach — the marginal cost of Bitcoin-anchored trust per action becomes negligible. It is not a cost barrier. It is a rounding error.

OpenTimestamps demonstrates this concretely. The protocol's public calendar servers batch thousands of timestamps into a single transaction, funded by the servers themselves, and offer the service for free to users.⁵⁶ The OriginStamp research documented a batching ratio of nearly 2,000:1 — nearly two million timestamps in approximately one thousand transactions.⁵⁷ The incremental cost per additional timestamp is effectively zero.

The economic model, then, is not "pay per transaction on Bitcoin." It is: "batch thousands of proofs, anchor once." The cost is borne not by individual agents but by the infrastructure operators — the trust service providers, the agent platforms, the enterprise compliance systems — who batch their clients' data and amortize the anchoring cost across all users. This is analogous to how internet infrastructure works: individual users do not pay for DNS queries or TCP/IP packets. The cost is embedded in the infrastructure and invisible to the end user.

There is one additional economic dimension worth noting. The Bitcoin base layer's throughput limitation — approximately seven transactions per second — is not a constraint on this architecture. Even at one transaction per batch per ten-minute block, the system can anchor 144 batches per day. At a batching ratio of 1,000,000:1, that is 144 million proofs per day from a *single* anchoring service. Multiple services can operate independently, each anchoring their own batches, scaling the system further without any coordination.

The throughput constraint of the base layer is, in this design, an asset rather than a liability. It is the constraint that forces the architecture to be efficient — to batch aggressively, to use block space parsimoniously — which in turn makes the system economically sustainable at any scale.

8.6 Strengthening the Evidentiary Stack: Completeness, Availability, and Economic Accountability

Section 8.2 established that Bitcoin-anchored audit trails provide tamper-proof evidence of what was recorded. It also acknowledged two limitations: the oracle problem (anchored records may not be truthful) and the completeness gap (not all events may have been recorded). These limitations cannot be eliminated at the substrate level — they are inherent to any commitment scheme. But they can be substantially narrowed through architectural measures at higher layers. This section describes four such measures, each addressing a specific gap in the evidentiary stack.

8.6.1 Multi-Agent Corroboration: Making Omission Detectable

Most economically meaningful agent interactions involve at least two parties. When Agent A transacts with Agent B, both agents can independently log the interaction and anchor their logs. If both parties' anchored records agree, the interaction is corroborated — verified by two independent sources. If the records disagree, the discrepancy is detectable and investigable. If one party's log omits an interaction that the other party's log records, the omission is evidenced.

This principle — bilateral or multilateral corroboration of shared events — does not require trust between the parties. It requires only that each party independently anchor its own log. The corroboration emerges from the structure of the interaction itself: any event that involves multiple agents produces multiple independent records, each independently anchored, each independently verifiable.

Corroboration does not guarantee completeness for purely internal agent actions (actions that affect only the agent itself and leave no external trace). But for the interactions that matter most for accountability — transactions, contract executions, data exchanges, authorization grants — corroboration makes selective omission detectable. An agent that omits a transaction from its log will find that the counterparty's log records it. The discrepancy between a present entry in one log and an absent entry in another is itself a powerful evidentiary signal.

8.6.2 Logging Cadence and Gap Detection

Continuous anchoring creates an expected cadence — a regular rhythm of Merkle root commitments to the Bitcoin blockchain. An agent that anchors its audit trail every block (approximately every ten minutes) establishes a verifiable heartbeat. Any interruption in this cadence — a period where no anchoring occurs — is observable by any party monitoring the blockchain.

Logging gaps do not prove malfeasance. An agent might go offline, reduce activity, or experience technical issues. But unexplained gaps during periods when other evidence suggests the agent was active (counterparty logs showing interactions, network traffic indicating communication) are evidentiary signals that demand explanation. The cadence creates an expectation; violations of the expectation shift the burden of proof.

This is analogous to how financial auditing works: regular, scheduled reporting creates a baseline against which anomalies can be detected. An enterprise that suddenly stops filing quarterly reports attracts regulatory scrutiny precisely because the expected cadence was broken. Continuous anchoring creates the same dynamic for agent audit trails.

8.6.3 Data Availability: Redundant Witnesses and the Transparency Log Model

Bitcoin anchors *proofs* (Merkle roots) but does not store the underlying *data* (log entries and Merkle trees). If the entity that maintains the off-chain data disappears, withholds it, or suffers data loss, the Merkle root on Bitcoin becomes unverifiable — you can prove something was anchored, but cannot produce the underlying evidence. This is the data availability problem, and it must be addressed at the architectural level.

The solution draws from the Certificate Transparency (CT) model, which has operated successfully at internet scale since 2013. In CT, multiple independent log operators maintain append-only logs of SSL/TLS certificates. Monitors observe the logs and detect anomalies. The key insight is *redundancy through independent witnesses*: no single operator's failure can destroy evidence, because multiple independent parties maintain copies.

For agent trust infrastructure, the data availability architecture operates as follows. When a batching service constructs a Merkle tree and anchors the root to Bitcoin, it simultaneously distributes the tree data — the log entries and their Merkle proofs — to multiple independent *witness nodes*. Witness nodes are operated by independent entities (compliance services, audit firms, industry consortia, open-source operators) who maintain copies of the anchored data. Any party seeking to verify an agent's audit trail can retrieve the data from any witness node, verify it against the Bitcoin-anchored Merkle root, and confirm integrity.

This architecture ensures that data availability does not depend on the cooperation of the agent's operator or the original batching service. Even if the operator withholds data, refuses to cooperate, or ceases to exist, the anchored records remain retrievable from independent witnesses. Non-delivery of data to witnesses — detectable by the witnesses themselves — can be penalized by exclusion from reputation systems or flagged as a compliance failure.

Economic incentives for witness operation include: compliance-service business models (enterprises pay for guaranteed evidence retention), insurance-market demand (insurers require data availability for claims adjudication), and regulatory mandates (the EU AI Act's traceability requirements create institutional demand for reliable evidence retention).

8.6.4 Economic Bonding: Reattaching Liability Through Verifiable Commitment

The trust substrate provides evidentiary infrastructure. But adversarial analysis has identified a deeper challenge: when agents operate at the far end of the accountability spectrum — multiple hops of delegation, cross-jurisdictional operation, unknown principals — even perfect evidence may not produce enforcement if no party with economic exposure can be identified.

Economic bonding addresses this by requiring agents (or their creators/deployers) to post verifiable collateral proportional to the agent's risk tier and operational scope. The mechanism operates as follows:

When an agent is deployed for high-stakes operations — financial transactions above a threshold, access to sensitive data, cross-jurisdictional commerce — its creator posts a bond (denominated in bitcoin, stablecoins, or tokenized real-world assets) to a publicly verifiable address. The bond is linked to the agent's Bitcoin-anchored identity certificate. The size of the bond reflects the risk tier: low-risk agents (customer service, internal automation) may require minimal or no bond; high-risk agents (financial trading, procurement, healthcare recommendations) require bonds proportional to the potential harm.

If the agent's anchored audit trail reveals misbehavior — or if a counterparty produces corroborating evidence of harm — the bond provides an economic enforcement surface. Slashing (automated forfeiture upon verified breach), insurance claims (bond serves as coverage for counterparty losses), and decentralized arbitration protocols can operate on the bond without requiring identification of a physical principal. The "missing body" problem is addressed not by finding a body but by ensuring that economic value is at risk — value that can be seized, redirected, or forfeited through verifiable on-chain mechanisms.

This approach draws from existing models. Ethereum's proof-of-stake uses economic slashing for validator misbehavior. DeFi protocols use collateralization for lending. The innovation here is applying economic bonding to the *agent trust* context — using Bitcoin-anchored identity and audit trails as the evidence layer, and posted bonds as the enforcement layer, creating a complete accountability stack that does not depend on identifying a physical human at the end of the chain.

Low-risk agents can operate permissionlessly with minimal or no bond — preserving the permissionless property of the substrate (Section 6.5). High-risk agents inherit economic accountability through their creators' bonds. The market determines the appropriate bond size for each risk tier, creating an economic equilibrium in which the cost of misbehavior (bond forfeiture) exceeds the potential benefit — the same incentive structure that makes most physical-world trust systems function.

This section has described how agent identity, action auditing, reputation, and the evidentiary stack can be built on the Bitcoin trust anchor using technology that already exists or is in active development. The architecture is layered, economically viable, and scales to the projected demands of the agentic economy. It does not require changes to the Bitcoin protocol. It does not require agents to "use Bitcoin" in the colloquial sense. It requires only that cryptographic proofs — compact, cheap, and mathematically verifiable — be periodically anchored to the most secure timestamping system ever created.

The remaining question is not technical feasibility. It is inevitability: how likely is it that this architecture — or something structurally similar to it — will actually be adopted? That is the subject of Section 9.

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9. The Inevitability Argument

This paper has presented a chain of reasoning: autonomous agents break the internet's trust model → the trust model cannot be repaired with existing tools → a new trust substrate is required → that substrate should satisfy six specific properties → Bitcoin is the strongest existing candidate that satisfies them. The chain is logically coherent, but logical coherence is not the same as inevitability. Many logically coherent outcomes never materialize.

This section offers an honest probabilistic assessment — not a prediction but a structured evaluation of how likely each link in the chain is to hold. The assessment proceeds from what is nearly certain to what is genuinely uncertain, because intellectual honesty about the weaker links strengthens the credibility of the stronger ones.

9.1 What Is Nearly Certain: Agents Will Become Autonomous Economic Actors

This is the foundation, and it is the strongest link in the chain.

The global AI agent market reached approximately \$7.8 billion in 2025 and is projected to exceed \$50 billion by 2030, reflecting a compound annual growth rate exceeding 45%.⁶⁰ Gartner forecasts that 40% of enterprise applications will embed task-specific AI agents by 2026, up from less than 5% in 2025.⁶¹ Approximately 85% of enterprises are expected to implement agents in at least one workflow by the end of 2026.⁶² These are not speculative projections from enthusiasts — they are consensus forecasts from the institutions that enterprises rely on for planning.

But the market data is less important than the competitive dynamics. The economic value of AI agents derives from their ability to act *without constant human supervision* — to operate at machine speed, in parallel, around the clock. An enterprise that constrains its agents to supervised, human-approved operations sacrifices the core value proposition: speed, scale, and continuous operation. An enterprise that grants its agents greater autonomy captures greater value — but assumes greater risk.

This creates a competitive ratchet. In any market where some participants grant their agents more autonomy and others grant less, the more autonomous agents outperform — faster execution, lower costs, broader coverage, 24/7 operation. Competitors must match or lose market share. The ratchet tightens in one direction only: toward greater agent autonomy. Any individual enterprise can choose to resist, but the market as a system cannot — because the enterprises that resist are systematically outcompeted by those that do not.

This is not novel. It is the same competitive dynamic that drove the adoption of the internet itself, of mobile computing, of cloud infrastructure. Each technology created a competitive asymmetry that made adoption effectively mandatory for any firm competing in global markets. AI agent autonomy follows the same pattern, with one critical difference: the stakes are higher, because the risk surface grows nonlinearly with agent autonomy.

The only plausible scenario in which agents do not become autonomous economic actors is coordinated global regulation that prohibits autonomous agent operation — a scenario that faces the same jurisdictional fragmentation problems identified in Section 4.3. Any nation that restricts agent autonomy while competitors do not will see its enterprises lose ground. Game theory predicts a race to permissiveness, not caution.

Assessment: Near certainty. The competitive dynamics are too powerful and the jurisdictional fragmentation too severe for coordinated restriction.

9.2 What Is Very Likely: Autonomous Agents Will Require a Neutral, Decentralized Trust Substrate

If agents become autonomous economic actors — which Section 9.1 argues is nearly certain — then the trust failures documented in Section 3 will intensify. The question is whether those failures can be addressed by centralized solutions or whether they require a neutral, decentralized substrate.

Section 4 analyzed the four main centralized approaches (more authentication, walled gardens, government identity, incremental adaptation) and identified structural failures in each. The most instructive is the walled garden analysis, because it addresses the strongest centralized alternative: protocols like Google's Agent2Agent (A2A) backed by major technology companies with the engineering resources and market position to potentially serve as trust intermediaries.

The structural problem is not engineering capability. It is geopolitics. The economic value of autonomous agents is maximized in *cross-boundary* interactions — agent-to-agent transactions that span organizations, platforms, jurisdictions, and regulatory regimes. These are precisely the interactions where centralized trust breaks down. A Google-mediated trust protocol works within the Google ecosystem. It works less well between Google and Microsoft ecosystems. It works poorly between American and Chinese ecosystems. It does not work at all in contexts where one party has legitimate reasons — competitive, regulatory, geopolitical — to refuse trust mediation by any single technology company.

The internet itself faced this problem in the 1990s, and the solution was the same: neutral protocols that no single entity controls. TCP/IP succeeded not because it was technically superior to every proprietary networking standard, but because it was neutral — no single company could weaponize it, revoke access to it, or extract rent from it. The same neutrality requirement applies to the trust layer of the agentic internet, for the same reasons.

The counterargument is that walled gardens could succeed by fragmenting the agent economy into competing ecosystems — a Google agent economy, a Microsoft agent economy, an Alibaba agent economy — each with internal trust but limited interoperability between them. This is possible but economically wasteful. It would replicate the pre-internet era of incompatible proprietary networks, sacrificing the enormous value of universal interoperability. Market pressure toward interoperability is strong. Interoperability requires neutral trust. And neutral trust, at the scale of the global agent economy, requires decentralization.

Assessment: Very likely. The geopolitical and competitive dynamics strongly favor neutral, decentralized trust infrastructure over centralized alternatives. The primary uncertainty is timing, not direction.

9.3 What Is Probable: The Trust Substrate Should Derive Security From Physics

This is the link where the argument becomes more distinctive and where reasonable disagreement begins.

Section 6 argued that the trust substrate should derive its security, to the greatest extent achievable, from physical constraints rather than institutional promises or mathematical assumptions. The reasoning proceeds from the documented failure patterns: every trust system that relied exclusively on institutional promises has eventually been broken under sufficient adversarial pressure. The SSL/TLS Certificate Authority system, designed with institutional trust at its core, suffered catastrophic failures (DigiNotar 2011, Symantec 2017). Government identity systems have been compromised repeatedly. Corporate databases are breached annually. The pattern is not that these systems are poorly designed — it is that promise-based security has inherent limits that accumulate over time.

Physical grounding — denominating the cost of attack in a physical resource rather than a digital asset or institutional promise — offers a qualitatively more robust foundation. But as Section 7.1 acknowledged, no

digital system achieves *unconditional* physical grounding. Bitcoin's proof-of-work is physics-grounded conditional on continued economic participation. The energy expenditure is real and irrecoverable, but the motivation to expend it is economic. This is a weaker claim than "thermodynamic guarantee" — but substantially stronger than any alternative offers.

The counterargument is that proof-of-stake systems or future cryptographic innovations could provide sufficient security without the energy expenditure of proof-of-work. This is not dismissable — cryptography is advancing, and it is possible that new mathematical primitives could provide guarantees that current systems cannot. But "possible" is not "probable." The known track record favors physical grounding: Bitcoin's proof-of-work has maintained unbroken security for sixteen years. No proof-of-stake system has operated at comparable scale for comparable duration against comparable adversarial pressure. The risk calculus for the root of trust of the global agent economy favors the system with the proven track record over the system with the theoretical promise.

An additional consideration: the dependency structure analyzed in Section 7.3. Proof-of-stake security has two layers of economic contingency (validator profitability and token value maintenance), while proof-of-work has one (mining profitability). For application-layer computation, this difference may be acceptable. For the foundational trust anchor, the engineering case for fewer and stronger dependencies is compelling, even if not logically compulsory.

Assessment: Probable, with legitimate room for disagreement. Physical grounding is the strongest engineering choice for a root of trust, but the possibility of alternative technical solutions providing sufficient security cannot be excluded. Confidence level: approximately 70-80%.

9.4 What Is the Strongest Candidate: Bitcoin's Competitive Position

Conditional on the previous three assessments — that agents will become autonomous economic actors, that they will require a neutral decentralized trust substrate, and that the substrate should derive security from physics — the question reduces to: which specific system?

Section 6's synthesis identified the requirements: natively digital, physically grounded, decentralized, immutable, temporally ordered, permissionless, and scarce. Section 7 evaluated Bitcoin against these requirements and found a complete match. But is Bitcoin the *only* possible system, or merely the best current candidate?

As of early 2026, no other operational system satisfies all six properties simultaneously. Ethereum provides decentralization and immutability but derives security from staked tokens (proof-of-stake), failing the physical grounding requirement. Permissioned blockchains provide immutability within their consortium but fail on decentralization and permissionlessness. Government-operated ledgers fail on decentralization, permissionlessness, and geopolitical neutrality. Corporate platforms fail on the same dimensions with additional commercial incentive risks.

Bitcoin's competitive position is reinforced by three factors that are difficult for any alternative to replicate:

Cumulative energy investment. Sixteen years of continuous proof-of-work mining have embedded hundreds of exajoules of energy into the Bitcoin blockchain. This energy is not recoverable — it represents an irreversible physical investment that makes the chain's history progressively harder to alter. Any competing proof-of-work chain would need to accumulate comparable energy investment to provide comparable security, which would take years and cost billions.

Network effects and institutional momentum. Bitcoin is the only cryptocurrency recognized in U.S. federal executive orders as a strategic reserve asset.⁶³ Multiple U.S. states have enacted or introduced

strategic Bitcoin reserve legislation.⁶⁴ The BITCOIN Act of 2025 proposes federal acquisition of up to one million BTC.⁶⁵ Institutional investors are projected to deploy hundreds of billions in capital into Bitcoin by end of 2026.⁶⁶ This institutional momentum creates a self-reinforcing cycle: institutional adoption increases Bitcoin's value, which increases miner revenue, which increases hashrate, which increases security, which makes Bitcoin more attractive to institutions.

The Lindy effect. Systems that have survived for a long time tend to survive longer. Bitcoin has operated without base-layer compromise for sixteen years — longer than any other decentralized system, under greater adversarial pressure, with higher economic stakes. Every year that passes without a successful attack increases confidence in the system's resilience, which increases the cost of switching to an alternative.

None of this means Bitcoin's position is permanent. A superior technology could emerge. But for a system that aspires to be the root of trust for the global agent economy, "proven and operating" is a stronger credential than "theoretically possible." The burden of proof lies with alternatives to demonstrate that they provide equivalent or superior guarantees, not with Bitcoin to demonstrate that no alternative could ever exist.

Assessment: Bitcoin is the strongest current candidate by a significant margin. Alternatives exist in theory but none currently match Bitcoin's combination of proven security, decentralization, institutional adoption, and cumulative energy investment.

9.5 The Reflexive Loop: Why Adoption Accelerates

The inevitability argument is strengthened by a feedback dynamic that distinguishes Bitcoin from most infrastructure technologies.

As described in Section 7.5, Bitcoin's architecture fuses the protocol with a scarce native asset. This creates a reflexive loop: adoption of Bitcoin's trust properties increases demand for BTC → increased demand increases miner revenue → increased revenue attracts more mining power → more mining power increases network security → stronger security makes Bitcoin's trust properties more valuable → which drives further adoption.

This loop operates in Bitcoin's existing use case (store of value / digital gold) and has been running for sixteen years. The agent trust use case adds a *second* reflexive loop on top of the first. If agents begin anchoring identity and audit trails to Bitcoin, the demand for block space increases. Increased block space demand increases transaction fees. Increased fees increase miner revenue. Increased revenue increases hashrate. Increased hashrate strengthens the trust anchor — making it more attractive for agents. The two loops compound.

Critically, reflexive loops are not merely additive — they are self-reinforcing and path-dependent. Once a reflexive system passes a critical threshold, the cost of displacing it grows faster than any competitor's ability to catch up. This is because the incumbent's security advantage grows with adoption, while a challenger must build comparable security from scratch. In the early stages, a challenger might compete. Past the threshold, the gap becomes structurally unbridgeable within any economically relevant timeframe.

Bitcoin may already be past this threshold for its monetary use case. If the agent trust use case begins to drive meaningful adoption, it would add a second self-reinforcing cycle to a system that is already deeply entrenched. The combination of two interlocking reflexive loops — one monetary, one infrastructural — would create a gravitational pull that is extraordinarily difficult to escape.

9.6 Honest Uncertainty: What Could Prevent This Outcome

Intellectual honesty requires acknowledging the scenarios in which this thesis does not materialize.

A cryptographic breakthrough that obsoletes proof-of-work. A new mathematical primitive that provides equivalent guarantees to proof-of-work without energy expenditure would undermine the core argument for physical grounding. This is theoretically possible but would represent a fundamental advance in both cryptography and our understanding of computational complexity. No such breakthrough is visible on the horizon, but the history of science teaches humility about predicting what is impossible.

Coordinated global prohibition of proof-of-work mining. If the world's major economies simultaneously banned Bitcoin mining on environmental or other grounds, the network's security could be compromised. This is possible but faces severe coordination problems: China already banned mining in 2021, and the hashrate simply migrated to other jurisdictions. A ban would need to be truly global — including jurisdictions with strong incentives to attract mining (low-cost energy, economic development, geopolitical positioning). History suggests this level of global coordination on economic policy is extraordinarily rare.

The agent revolution stalling. If autonomous agents do not achieve the scale and autonomy projected — if regulation, technical limitations, or public backlash significantly slows adoption — then the trust crisis described in Section 3 may not reach the severity that demands a new trust substrate. This is the most plausible counterargument to the thesis. However, even a significant slowdown in agent adoption merely delays the problem; it does not eliminate it. As long as the trajectory is toward greater agent autonomy, the trust substrate question must eventually be addressed.

A "good enough" centralized solution. The most pragmatic counterargument: perhaps a coalition of major technology companies, backed by government regulation, creates a centralized or semi-centralized trust infrastructure that is "good enough" for most use cases. This would not satisfy the theoretical requirements of Section 6 — it would have single points of failure, jurisdictional limitations, and censorship risks. But most infrastructure in the real world is "good enough," not theoretically optimal. The history of the internet includes many cases where the technically superior solution lost to the more convenient one.

This is the scenario most worth monitoring. It is the one most likely to prevent Bitcoin from becoming the universal trust anchor — not because a better decentralized alternative emerges, but because a worse centralized alternative proves sufficient for the majority of use cases. The counterargument to this scenario is the geopolitical analysis of Section 4.2: any centralized system is inherently unacceptable to some subset of global participants, and the value of agent commerce depends on universal participation.

Assessment of overall thesis: The chain from "agents will become autonomous" to "they will need a trust substrate" to "Bitcoin is the strongest candidate" reflects a high-probability outcome, with the weakest links being the requirement for physical grounding (where alternatives are theoretically possible) and the speed of adoption (where "good enough" centralized solutions could delay the transition). The gravitational pull toward this outcome is strong. It is not inevitable. But the burden of argument now lies with those who claim it will *not* happen, not with those who claim it will.

9.7 Alternative Substrates: Engaging the Strongest Counter-Proposals

Intellectual honesty requires engaging not only with scenarios where the problem doesn't materialize (Section 9.6), but with scenarios where the problem is real and the solution is different. Two alternative substrate proposals merit serious analysis: they accept the problem statement of this paper but arrive at different architectures.

The Transparency-Log Internet. The strongest practical alternative generalizes Certificate Transparency (CT), a system that has operated successfully at internet scale since 2013. In this model, many independent log operators run append-only Merkle logs for agent events — identity issuance, key rotation, action receipts. Logs publish signed tree heads on a regular cadence. Tree heads are cross-logged into each other (a "log-of-logs" graph) to prevent equivocation — the ability to present different versions of the log to different observers. Clients and monitors use gossip protocols to detect split views and missing consistency proofs. Timestamp bounds come from multi-source time beacons rather than a cryptocurrency chain. Availability is improved via redundant witness retention.

This proposal has real merit, and elements of it are incorporated into this paper's architecture. Section 8.6.3's data availability model draws directly from the CT approach — redundant witnesses, independent log operators, gossip-based consistency checking. These are proven engineering patterns that belong in any serious trust infrastructure.

Where the Transparency-Log Internet falls short as a *standalone substrate* — as a replacement for, rather than complement to, Bitcoin anchoring — is in two properties from Section 6. First, **physical grounding**: CT-style logs rely on the honesty of log operators. If a majority of operators collude (or are coerced by a state actor), they can present consistent but fabricated tree heads. The security is denominated in the collective integrity of institutional operators — which is promise-based trust, the very dependency the substrate should minimize. Bitcoin's proof-of-work provides a qualitatively different guarantee: the cost of fabricating history is denominated in energy expenditure, not institutional honesty. Second, **a single consistent global ordering**: CT logs provide ordering within each log but not a canonical global ordering across all logs. Cross-logging and gossip detect equivocation after the fact but do not prevent it — they create eventual consistency, not the strict, physics-enforced temporal ordering that Bitcoin's blockchain provides. For disputes that hinge on which event came first across different logs, this gap matters.

The most productive framing is complementary: the Transparency-Log model describes an excellent Layer 2/3 architecture *built on top of* Bitcoin's Layer 0 anchoring. Use Bitcoin for the global ordering primitive and the physics-grounded immutability guarantee. Use CT-style redundant witnesses for data availability and equivocation detection. This combination captures the strengths of both approaches while compensating for their respective limitations.

Entropy-Ordered Gossip Networks. A more speculative alternative proposes a peer-to-peer gossip network where every high-stakes agent action is accompanied by a verifiable sample of physical entropy drawn from distributed entropy beacons — radioactive decay detectors, atmospheric noise receivers, satellite-based quantum random number generators — operated by independent entities in multiple jurisdictions. The entropy sample serves as both a physical-grounding mechanism and a non-forgeable ordering token: ordering is established by the physical propagation delays of the gossip network and by the earliest verifiable entropy sample that could have been observed. Scarcity is introduced by requiring each commitment to consume a small, verifiable amount of the entropy beacon's rate-limited output.

This is the most intellectually interesting alternative because it genuinely attempts to satisfy all six Section 6 properties through a novel mechanism — importing physics not through energy expenditure (proof-of-work) but through thermodynamic randomness (entropy). It avoids the economic contingency that all three critiques of Bitcoin's physics-grounding identify: entropy beacons produce entropy regardless of any token's market price.

Where it falls short is in three practical dimensions that prevent deployment today. First, zero-knowledge proofs for arbitrary agent execution — proving not just *that* an action occurred but *what* the action was, without revealing the agent's internal state — do not exist at the generality required. Current ZK proof systems work for specific, well-defined computations; proving the execution trace of an arbitrary neural network agent is an unsolved research problem. Second, entropy beacons require trusted hardware —

someone must manufacture, calibrate, and certify the radioactive decay detectors or quantum random number generators. This reintroduces institutional trust in hardware manufacturers, the same dependency that undermines TEE-based proposals (though distributed across multiple independent manufacturers, partially mitigating the risk). Third, gossip networks face fundamental consistency and finality challenges — in the absence of a global ordering mechanism (which is what the entropy is supposed to provide), split views and ordering disputes require resolution mechanisms that inevitably reintroduce either centralization or the eventual-consistency limitations of the Transparency-Log model.

Entropy-ordered gossip is a research direction worth monitoring, not a deployable infrastructure. If the underlying technical challenges — general-purpose ZK proofs for agent execution, truly decentralized entropy beacon networks, gossip consistency with strong finality guarantees — are solved, the resulting system could provide a physics-grounded substrate without the economic contingency of proof-of-work. That day is years away. The trust infrastructure gap is now.

The broader insight from alternative analysis. The two strongest counter-proposals reinforce rather than undermine this paper's core thesis. The Transparency-Log Internet accepts the need for the primitives Bitcoin provides (immutability, ordering, tamper-evidence) and proposes a different mechanism — one that, on analysis, works best as a complementary layer rather than a replacement. Entropy-ordered gossip networks accept the need for physics-grounded security and attempt a different mechanism — one that remains theoretical. Both alternatives implicitly validate the problem specification of Section 6. The disagreement is about which existing or proposed system best satisfies it — and in that comparison, Bitcoin's combination of sixteen years of proven operation, cumulative energy investment, institutional adoption, and architectural soundness remains the strongest available candidate.

9.8 Why Now: The Timing Is Not Coincidence

A skeptic might ask: "Bitcoin has existed since 2009. Proof-of-work's properties have been known for sixteen years. Why does this suddenly matter?"

The answer is that the problem Bitcoin solves — the absence of a physics-grounded trust substrate in cyberspace — was not concretely visible until now. The theoretical framework existed. Jason Lowery's *Softwar* (2023) articulated the case for proof-of-work as a mechanism for projecting physical power into cyberspace. But in 2022-2023, the thesis felt abstract. The actors in cyberspace were still overwhelmingly human. The internet's institutional trust model was strained but functional. The abstraction chain — every digital identity tracing back to a human body — was fraying but intact.

The emergence of autonomous AI agents changed this. Not gradually, but as a phase change. For the first time, cyberspace contains actors with no physical bodies, no jurisdictions, no human accountability. The abstraction chain doesn't just fray for these actors — it degrades along a spectrum from attenuated to absent, with the most economically valuable interactions occurring precisely where the chain is most degraded. The trust model that was adequate when every actor was ultimately human is architecturally inadequate when actors are machines operating at machine speed across jurisdictional boundaries.

The timing is causal, not coincidental. Agents did not merely make the trust problem bigger. They made it *exist*. Before agents, the internet's trust model was adequate because the actors it governed were the same kind of actors it was designed for — humans with bodies. The trust substrate was a bandage, but the patient wasn't bleeding. Now the patient is hemorrhaging, and the bandage is visibly failing.

This is why Lowery's thesis, which felt premature in 2023, now reads as prescient. The theoretical necessity he identified — that proof-of-work is the mechanism by which laws of physics are imported into cyberspace — is now a practical urgency. The autonomous digital beings that make the theory concrete have arrived. The trust architecture that makes the solution necessary has begun to fail. The window

between "the problem is visible" and "the problem is catastrophic" is the window in which infrastructure must be built.

We are in that window now.

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10. Implications

If the thesis of this paper is correct — that the agentic internet requires a trust substrate rooted in physics, and that Bitcoin is the strongest existing system that provides one — then the implications extend far beyond cryptocurrency markets. They reach into how software is built, how enterprises deploy AI, how the Bitcoin ecosystem evolves, and how governments regulate artificial intelligence.

This section maps those implications for four audiences, in order of urgency.

10.1 For Builders: The Infrastructure Gap That Exists Right Now

The trust substrate exists. The infrastructure built on it does not.

Bitcoin provides the raw primitives: timestamping, ordering, immutability, scarcity. But the layers between these primitives and the enterprise customer — the identity protocols, the audit anchoring services, the reputation computation engines, the SDKs, the compliance dashboards, the agent platform integrations — are almost entirely unbuilt. The gap between "Bitcoin can serve as a trust anchor for autonomous agents" and "an enterprise can deploy Bitcoin-anchored agent auditing in production" is a gap of engineering, product design, and go-to-market execution.

This gap represents the defining infrastructure opportunity of the agentic era. Consider the scope of what must be built:

Agent identity infrastructure. Protocols for generating, anchoring, updating, and revoking Bitcoin-anchored agent certificates. Standards for certificate formats that are interoperable across agent platforms (LangChain, CrewAI, AutoGPT, enterprise custom frameworks). Discovery mechanisms that allow one agent to verify another's identity without a central registry. Key management solutions that handle the reality of agents operating across cloud environments, edge devices, and hybrid deployments.

Audit anchoring services. Managed services that handle the Merkle tree construction, batching, and Bitcoin anchoring described in Section 8.2 — abstracting the complexity so that an enterprise can integrate audit anchoring with a single API call. Compliance-grade implementations that satisfy EU AI Act requirements for immutable audit trails and traceability.⁶⁷ Verification tools that allow regulators, auditors, and counterparties to independently check audit trail integrity without specialized blockchain knowledge.

Reputation computation. Algorithms that derive trust scores from Bitcoin-anchored behavioral data — weighting identity age, audit trail consistency, transaction patterns, interaction graph quality, and anomaly indicators. Privacy-preserving implementations that allow reputation to be evaluated without exposing the underlying behavioral data. Reputation APIs that agent platforms can query in real time before executing cross-agent transactions.

Developer tooling. SDKs in every major language (Python, TypeScript, Go, Rust) that make Bitcoin trust anchoring as simple as adding a dependency. Integration plugins for major agent orchestration frameworks. Testing and simulation environments where developers can prototype trust-anchored agent systems without mainnet costs.

Compliance and monitoring. Enterprise dashboards that provide real-time visibility into agent audit trails, identity status, and reputation metrics. Alert systems that flag anomalies — gaps in audit trails, unusual identity modifications, reputation degradation. Reporting tools that generate EU AI Act compliance documentation from Bitcoin-anchored records.

The strategic window for building this infrastructure is narrow and favorable. The problem is visible enough to attract early customers — enterprises deploying agents are beginning to grapple with audit, identity, and compliance challenges. But it is not yet obvious enough to attract heavy competition from large incumbents, whose natural instinct is to solve trust problems with centralized platforms (their existing business model). The window will close as the problem becomes acute: at that point, well-funded incumbents will enter, and the advantage will shift from first-movers to those with distribution. The builders who establish the infrastructure layer now — before the crisis is obvious — will define the architecture of the agentic internet's trust layer.

10.2 For Enterprises: Preparing for Agent Compliance

Enterprises deploying AI agents face a regulatory environment that is tightening faster than most compliance teams realize.

The EU AI Act's requirements for high-risk AI systems become enforceable on August 2, 2026 — less than six months from this writing.⁶⁷ The requirements include mandatory risk management systems, technical documentation, automated logging of AI system activity with sufficient traceability, human oversight mechanisms, and conformity assessments before deployment. Penalties for non-compliance reach €35 million or 7% of global annual revenue, whichever is higher.⁶⁸ The regulation applies extraterritorially: any organization whose AI systems affect EU residents must comply, regardless of where the organization is headquartered.

The United States is moving in the same direction, though through a patchwork of state-level legislation rather than a single federal framework. Colorado's AI Act, effective February 2026, regulates high-risk AI systems in consequential decisions including employment, housing, education, and financial services.⁶⁹ Other states are advancing similar legislation. The trajectory is clear even if the federal timeline remains uncertain.

These regulations share a common requirement that maps directly to the architecture described in Section 8: *verifiable, immutable audit trails* for AI system activity. The EU AI Act specifically requires logging capabilities that enable traceability and monitoring of system behavior. It requires documentation that can be provided to authorities to demonstrate compliance. It requires that these records be resistant to tampering.

Most enterprises today address these requirements with traditional database logging — records stored in enterprise databases, protected by access controls, maintained by database administrators. This is promise-based immutability. It satisfies the letter of current guidance but not its spirit, because the records can be altered by anyone with administrative access, and their integrity depends on the trustworthiness of the organization maintaining them.

As agent-driven incidents increase (Section 3) and regulatory scrutiny intensifies, the gap between promise-based and constraint-based immutability will become a material compliance risk. An enterprise that can demonstrate its agent audit trails are anchored to the Bitcoin blockchain — independently verifiable by any third party, mathematically proven to be unaltered — has a categorically stronger compliance position than one relying on internal database logs. The former can prove its records are intact. The latter can only assert it.

The practical recommendation is straightforward: enterprises deploying AI agents in any capacity that might fall under high-risk classification should begin evaluating Bitcoin-anchored audit infrastructure now, before the August 2026 enforcement deadline. The integration cost is modest (Section 8.5 demonstrated negligible per-action costs). The compliance benefit is substantial. And the reputational benefit — being able to demonstrate to customers, partners, and regulators that your agent audit trails are anchored to the most secure timestamping system ever created — is a competitive advantage that early adopters will exploit.

10.3 For the Bitcoin Ecosystem: The Largest New Use Case Since Store of Value

Bitcoin's primary narrative for the past decade has been "digital gold" — a store of value secured by proof-of-work, with a fixed supply cap of 21 million coins. This narrative has driven Bitcoin to a market capitalization exceeding \$1.4 trillion as of early 2026, attracted institutional investors, sovereign wealth funds, and — in an unprecedented development — the United States government itself, which established a Strategic Bitcoin Reserve by executive order in March 2025.⁷⁰

The agent trust use case described in this paper represents a second, potentially larger narrative: Bitcoin as the trust infrastructure of the agentic internet. These two narratives do not compete. They compound.

The store-of-value narrative drives demand for BTC as an asset, which increases miner revenue, which

increases hashrate, which strengthens network security. The trust infrastructure narrative drives demand for Bitcoin block space (for anchoring), which increases transaction fees, which increases miner revenue, which further strengthens security. The two revenue streams — asset appreciation and block space demand — flow to the same miners, reinforcing the same security model. An investor holding BTC for store-of-value purposes benefits from the additional security provided by trust infrastructure demand, and vice versa.

This has specific implications for Bitcoin's long-term security model. Bitcoin's block reward — the newly minted BTC that compensates miners — halves approximately every four years and will eventually reach zero. The network's security must eventually be sustained entirely by transaction fees. The agent trust use case provides a structurally recurring source of fee revenue: enterprises anchoring agent audit trails, identity services batching certificates, reputation systems querying the chain. Unlike speculative trading volume (which fluctuates with market sentiment), compliance-driven anchoring generates consistent, predictable demand for block space — the kind of demand that can sustain mining economics long after block rewards diminish.

For Bitcoin developers, the implication is that the ecosystem's highest-value development work over the coming decade may not be in financial applications (exchanges, custody, lending) but in trust infrastructure: the protocols, tools, and platforms that make Bitcoin-anchored agent identity, auditing, and reputation accessible to enterprise customers. This is a shift in emphasis, not direction — it builds on Bitcoin's existing properties rather than requiring protocol changes.

For Bitcoin investors, the implication is that the addressable market for BTC extends beyond the store-of-value thesis (competing with gold's \$15+ trillion market cap) to include the trust infrastructure thesis (undergirding every agent transaction, identity, and audit trail in the global agent economy). If the AI agent market reaches \$50 billion by 2030⁷¹, and agent trust infrastructure captures even a fraction of that market, the demand for Bitcoin block space and BTC-denominated services adds a new, independent driver of network value.

10.4 For Policymakers: Regulation That Enables Rather Than Restricts

Policymakers face a consequential choice in how they regulate autonomous AI agents. The choice is not between regulation and no regulation — that debate is settled; regulation is coming. The choice is between two regulatory architectures that lead to very different outcomes.

Architecture 1: Mandate centralization. Require agents to register with government-approved identity providers. Require all agent activity to be logged in government-accessible databases. Require agent operators to obtain licenses from centralized authorities. This architecture is familiar — it mirrors how governments regulate financial services, telecommunications, and pharmaceuticals. It is the default instinct of regulatory bodies.

The problem: this architecture creates the walled gardens analyzed in Section 4.2. It fragments the agent economy along jurisdictional lines. It creates single points of failure that adversaries can target. It grants governments the power to weaponize trust infrastructure for geopolitical purposes. It disadvantages enterprises in jurisdictions with restrictive regulators. And it does not solve the fundamental problem — that centralized identity systems can be gamed by sufficiently capable agents, as Section 3 documented extensively.

Architecture 2: Mandate verifiability. Require that agent activity be recorded in verifiable, immutable audit trails — but do not mandate *where* those trails are stored or *who* maintains them. Require that agent identities be provable — but do not mandate which identity system issues them. Require that compliance evidence be independently auditable — but do not mandate which auditing platform checks it.

This architecture is technology-neutral. It specifies *outcomes* (verifiable records, provable identities, auditable compliance) rather than *mechanisms* (specific databases, specific platforms, specific registries). It allows the market to determine which trust infrastructure best satisfies the requirements — and the analysis of this paper suggests that Bitcoin-anchored infrastructure will emerge as the strongest candidate, precisely because it satisfies the verifiability and immutability requirements more rigorously than any centralized alternative.

The policy principle is simple: **mandate auditability, not centralization. Require provability, not permission.**

The EU AI Act, to its credit, moves partially in this direction. Its requirements for immutable audit trails, traceability, and documentation are outcome-oriented — they specify what must be demonstrated, not which specific technology must demonstrate it. This creates regulatory demand for exactly the kind of infrastructure described in Section 8. The more explicitly regulators adopt outcome-based requirements, the more naturally Bitcoin-anchored trust infrastructure emerges as the compliance solution.

The risk is that regulatory bodies, under pressure to act quickly as agent-driven incidents increase, default to Architecture 1 — mandating centralized identity registries, platform-specific compliance, and government-controlled audit databases. This path is faster to implement but architecturally fragile, geopolitically divisive, and ultimately insufficient against machine-speed adversaries operating across jurisdictions.

The stakes of this choice are high. The regulatory architecture for autonomous agents will shape the structure of the agentic internet for decades — just as the regulatory decisions of the 1990s (the U.S. policy of minimal internet regulation, the EU's embrace of open standards) shaped the structure of the internet we have today. Policymakers who understand the trust substrate problem described in this paper have the opportunity to create regulation that *accelerates* the development of robust, decentralized, physics-grounded trust infrastructure — rather than regulation that entrenches fragile centralized alternatives.

10.5 The Deeper Implication: Incentive Realignment at Civilizational Scale

There is a final implication that extends beyond any single audience — one that addresses the full consequence of what a physics-grounded trust substrate means for human and machine behavior at scale.

The internet, as it exists today, operates under a broken incentive structure. Fraud is cheap. Fabrication is free. Deception scales at machine speed while verification crawls at human speed. The rational economic actor in this environment is the one who exploits the asymmetry — who generates fake identities, fabricates reputations, executes unauthorized transactions — because the cost of deception is negligible and the probability of accountability is low.

A trust substrate grounded in physics reverses this asymmetry.

When every agent identity has a verifiable, immutable provenance record anchored to the Bitcoin blockchain, fabricating an identity requires expending real resources — and the fabrication leaves detectable traces. When every high-stakes action is logged in a tamper-proof audit trail, misconduct becomes permanently attributable — not through institutional investigation but through mathematical proof. When reputation is derived from verifiable behavioral history rather than self-reported credentials, honesty becomes the optimal strategy because honest behavior accumulates verifiable trust capital while dishonest behavior either fails to accumulate it or destroys it.

This is not a moral argument. It is a game-theoretic one. In an environment where deception is cheap and detection is expensive, the equilibrium favors deception. In an environment where deception is expensive

and detection is cheap, the equilibrium favors honesty. A physics-grounded trust substrate shifts the cost structure from the former to the latter — not by making fraud impossible, but by making fraud *economically irrational* at scale.

The second-order consequences of this shift cascade through every domain the paper has examined. In finance, the cost of agent-driven fraud becomes bounded by the cost of creating verifiable false histories — which, on a proof-of-work-anchored system, is denominated in real energy expenditure rather than free computation. In commerce, fake storefronts and fabricated reviews become detectable through reputation analysis against immutable behavioral records. In identity, synthetic personas become distinguishable from genuine ones through the depth and consistency of their Bitcoin-anchored history. In infrastructure, cascading agent failures become traceable through audit trails that cannot be retroactively altered.

The third-order consequence is perhaps the most profound. When the cost of deception rises and the cost of verification falls, the *efficiency* of the entire system improves. Transactions that currently require extensive due diligence (because counterparty trust cannot be verified) can proceed with minimal friction (because counterparty trust *can* be verified, mathematically, in milliseconds). The economic gains from reduced friction — in a global economy where billions of agent-to-agent transactions will occur daily — are difficult to overstate. Every hour spent on due diligence, every dollar spent on fraud investigation, every transaction delayed by verification uncertainty represents economic waste that a functioning trust substrate eliminates.

This is the positive reflexive loop that extends beyond Bitcoin or agents or any single technology: a trust substrate that makes honesty the winning strategy creates an environment in which more actors choose honesty, which makes the environment more trustworthy, which reduces friction, which increases economic productivity, which creates more value, which attracts more participation, which further strengthens the trust substrate.

The internet gave humanity a communication substrate that connected every person on earth. What it did not provide — what it was never designed to provide — was a trust substrate. For three decades, this absence was tolerable because the actors in cyberspace were humans whose trust was ultimately grounded in their physical bodies. That era is ending. The actors are no longer exclusively human. The trust must be grounded in something else — something as fundamental, as reliable, and as universal as physics itself.

The infrastructure that provides this grounding is the generational building opportunity of the next decade.

References — Section 10

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11. Conclusion

The Argument

For three decades, the internet has operated without a trust layer. It has communication protocols (TCP/IP), naming protocols (DNS), and encryption protocols (TLS) — but no trust protocol. No foundational, universal mechanism by which one entity can verify the identity, history, or trustworthiness of another without relying on a human institution to vouch for them.

This absence was tolerable because every actor in cyberspace was, ultimately, a human with a physical body. Digital identities traced back to government-issued documents. Online reputations were constrained by the scarcity of human time. Accountability was enforceable because people had physical addresses, operated within legal jurisdictions, and could not outrun the consequences of their actions. The internet's trust model was a bandage — institutional trust borrowed from the physical world and stretched across digital infrastructure — but it held, because the actors it governed were the same kind of actors it was designed for.

That era is ending.

Autonomous AI agents are entering the global economy as independent actors — executing transactions, negotiating contracts, managing infrastructure, accessing data — at machine speed, across jurisdictions, without constant human oversight. These agents have no bodies. No passports. No physical addresses. No vulnerability to the mechanisms by which human societies have enforced accountability for millennia. When the abstraction chain that connects every digital identity to a physical human body terminates not at a person but at a process — instantiated in milliseconds, replicated without limit, terminated without consequence — the entire trust architecture of the internet breaks. Not at the edges. At the foundation.

This paper has demonstrated, through documented incidents and structural analysis, that the break is already underway. Deepfake-driven fraud has cost hundreds of millions of dollars. Synthetic identities have infiltrated Fortune 500 companies. Compromised agents have cascaded failures through enterprise supply chains. Autonomous AI has conducted espionage campaigns against dozens of targets with minimal human involvement. Malicious skills have exfiltrated data from the most widely deployed agent platform in the world. These are not projections. They are events of 2024 and 2025.

The paper examined the four most commonly proposed solutions — more authentication, walled gardens, government identity, incremental adaptation — and found that each fails for the same structural reason: each attempts to extend trust mechanisms designed for human actors to govern non-human actors. The category of solutions is wrong, not merely the implementation.

What is required is a new category: a trust substrate. A foundational layer — analogous to TCP/IP for communication or DNS for naming — that provides primitive trust operations (identity, temporal ordering, immutability, scarcity) upon which higher-level systems can build. This substrate must be natively digital

(accessible to entities without physical bodies) and physically grounded to the greatest extent achievable (deriving its core security from the laws of physics rather than from institutional promises alone). It must be decentralized, immutable, temporally ordered, permissionless, and scarce.

The strongest existing candidate that satisfies these requirements is Bitcoin. Its proof-of-work mechanism converts electricity — a physical resource constrained by the laws of thermodynamics — into cryptographic proof, importing the scarcity, irreversibility, and temporal ordering of the physical world into cyberspace. The physical grounding is conditional on continued economic participation — miners mine for profit, not for physics — but this economic contingency is simpler, more robust, and more externally anchored than any alternative. Sixteen years of continuous operation without a successful base-layer attack, sustained against the most economically incentivized adversarial pressure of any digital system in history, provide the empirical foundation that theory alone cannot.

The architecture for building agent trust on this substrate already exists in prototype. Bitcoin-anchored identity certificates provide verifiable, tamper-proof agent provenance. Merkle tree batching with Bitcoin anchoring — operational since 2016 through OpenTimestamps — enables immutable audit trails at negligible per-action cost. Derived reputation systems, computed from publicly verifiable behavioral records, enable trust between agents that have never interacted. Multi-agent corroboration and redundant witness nodes narrow the completeness gap — making omission detectable even when it cannot be prevented. Economic bonding reattaches liability to agents operating at the far end of the accountability spectrum — ensuring that economic value is at risk even when no physical principal can be identified. The substrate does not, by itself, restore accountability — it provides tamper-proof records of what was recorded (eliminating post-creation tampering), architectural measures for completeness verification (making omission detectable), and economic enforcement surfaces (ensuring consequences for misbehavior). Together with the existing institutional machinery of courts, regulators, and contracts, these form a complete accountability stack for the agentic era. The technology is proven. What is missing is integration — the assembly of existing components into a coherent trust infrastructure for the agentic internet.

The Thesis

The agentic internet requires a trust anchor rooted in physics. Bitcoin is the strongest existing system that provides one. The infrastructure connecting the two is the defining build opportunity of this decade.

The Window

The window between "the problem is visible" and "the problem is catastrophic" is the window in which infrastructure must be built. The trust failures documented in this paper are early signals — the first cracks in an architecture that was never designed for the forces now acting upon it. The incidents will intensify. The regulatory response will accelerate. The demand for verifiable, immutable, physics-grounded trust infrastructure will grow from niche concern to existential requirement.

Those who build this infrastructure now — before the crisis is obvious — will define the trust architecture of the agentic internet. Those who wait will build on foundations laid by others.

The internet connected humanity. The agentic internet will test whether that connection can survive without trust. The answer depends on what we build next.
